

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 1                 |        |        |        |        |        |        |
| DLT rate                   | 0.30   | 0.35   | 0.40   | 0.45   | 0.50   |        |
| Efficacy rate              | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |        |
| Utility 2                  | -0.2   | -0.3   | -0.3   | -0.4   | -0.4   |        |
| Utility 3                  | 31     | 29.6   | 28.2   | 26.8   | 25.4   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.9    | 2.3    | 0.3    | 0.1    | 0      | 92.4   |
| U2 - cutoff = 0.85         | 13.2   | 4      | 0.5    | 0.1    | 0      | 82.2   |
| U2 - cutoff = 0.95         | 25.9   | 10.6   | 0.2    | 0      | 0      | 63.3   |
| U3 - cutoff = 0.75         | 5.7    | 1.8    | 0.2    | 0      | 0      | 92.3   |
| U3 - cutoff = 0.85         | 14.4   | 3.4    | 0.4    | 0      | 0      | 81.8   |
| U3 - cutoff = 0.95         | 28.9   | 7.8    | 0.1    | 0      | 0      | 63.2   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 8.1    | 2.6    | 1.2    | 0.3    | 0.1    | -      |
| U2 - cutoff = 0.85         | 11.5   | 2.9    | 1.3    | 0.3    | 0.1    | -      |
| U2 - cutoff = 0.95         | 16.2   | 3.7    | 1.3    | 0.3    | 0.1    | -      |
| U3 - cutoff = 0.75         | 8.1    | 2.6    | 1.2    | 0.3    | 0.1    | -      |
| U3 - cutoff = 0.85         | 11.5   | 2.9    | 1.3    | 0.4    | 0.1    | -      |
| U3 - cutoff = 0.95         | 16.3   | 3.7    | 1.3    | 0.3    | 0.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 2.1    | 1.4    | 0.4    | 0.2    | 0      | -      |
| U2 - cutoff = 0.85         | 3.4    | 1.6    | 0.5    | 0.2    | 0      | -      |
| U2 - cutoff = 0.95         | 0.5    | 1.9    | 0.4    | 0.2    | 0      | -      |
| U3 - cutoff = 0.75         | 2.1    | 1.4    | 0.4    | 0.2    | 0      | -      |
| U3 - cutoff = 0.85         | 3.4    | 1.6    | 0.5    | 0.2    | 0      | -      |
| U3 - cutoff = 0.95         | 0.5    | 1.9    | 0.4    | 0.2    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 439.6  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 573.9  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1030.7 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 440.9  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 573.1  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1031.9 |
|                            |        |        |        |        |        |        |
| Scenario 2                 |        |        |        |        |        |        |
| DLT rate                   | 0.15   | 0.20   | 0.30   | 0.35   | 0.45   |        |
| Efficacy rate              | 0.20   | 0.25   | 0.50   | 0.50   | 0.50   |        |
| Utility 2                  | 0.1    | 0.1    | 0.2    | 0.1    | 0.1    |        |
| Utility 3                  | 46     | 47     | 58     | 56     | 52     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 32.3   | 18.2   | 30     | 6.8    | 1.4    | 11.3   |
| U2 - cutoff = 0.85         | 37.5   | 20.3   | 30.3   | 6.7    | 1.1    | 4.1    |
| U2 - cutoff = 0.95         | 35.3   | 24.1   | 29.8   | 7.6    | 1.2    | 2      |
| U3 - cutoff = 0.75         | 34.9   | 19.8   | 27.8   | 5.9    | 0.5    | 11.1   |
| U3 - cutoff = 0.85         | 40.5   | 20.5   | 28.2   | 6      | 0.7    | 4.1    |
| U3 - cutoff = 0.95         | 39     | 24.4   | 26.6   | 7.1    | 0.9    | 2      |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 11.3   | 5.5    | 6.8    | 3.2    | 1.2    | -      |
| U2 - cutoff = 0.85         | 12.5   | 5.6    | 6.9    | 3.1    | 1.1    | -      |
| U2 - cutoff = 0.95         | 12.7   | 6.2    | 6.7    | 3      | 1.1    | -      |
| U3 - cutoff = 0.75         | 11.3   | 5.5    | 6.9    | 3.1    | 1.2    | -      |
| U3 - cutoff = 0.85         | 12.5   | 5.6    | 6.9    | 3.1    | 1.2    | -      |
| U3 - cutoff = 0.95         | 12.8   | 6.1    | 6.6    | 3.1    | 1.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 3.2    | 2.6    | 1.9    | 0.9    | 0.3    | -      |
| U2 - cutoff = 0.85         | 3.7    | 2.6    | 1.9    | 0.9    | 0.3    | -      |
| U2 - cutoff = 0.95         | 0.4    | 2.6    | 1.2    | 0.7    | 0.2    | -      |
| U3 - cutoff = 0.75         | 3.2    | 2.6    | 1.9    | 0.9    | 0.3    | -      |
| U3 - cutoff = 0.85         | 3.7    | 2.6    | 1.9    | 0.9    | 0.3    | -      |
| U3 - cutoff = 0.95         | 0.4    | 2.6    | 1.2    | 0.7    | 0.2    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1052.6 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1094.3 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1358.5 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1053.2 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1095.5 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1358.5 |

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 3                 |        |        |        |        |        |        |
| DLT rate                   | 0.15   | 0.20   | 0.30   | 0.35   | 0.45   |        |
| Efficacy rate              | 0.10   | 0.40   | 0.40   | 0.40   | 0.40   |        |
| Utility 2                  | -0.1   | 0.2    | 0.1    | 0.1    | -0.1   |        |
| Utility 3                  | 40     | 56     | 52     | 50     | 46     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 9.5    | 37.4   | 18.5   | 5.9    | 0.8    | 27.9   |
| U2 - cutoff = 0.85         | 15.2   | 40.4   | 19.1   | 6      | 0.6    | 18.7   |
| U2 - cutoff = 0.95         | 19.2   | 47.4   | 16.2   | 5.4    | 0.6    | 11.2   |
| U3 - cutoff = 0.75         | 10.8   | 39.3   | 16.9   | 4.6    | 0.5    | 27.9   |
| U3 - cutoff = 0.85         | 16.4   | 43.2   | 16.9   | 4.4    | 0.5    | 18.6   |
| U3 - cutoff = 0.95         | 19.9   | 49.2   | 14.8   | 4.2    | 0.5    | 11.5   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 7.3    | 7.7    | 5.7    | 3      | 1.1    | -      |
| U2 - cutoff = 0.85         | 9.4    | 7.8    | 5.7    | 3      | 1.1    | -      |
| U2 - cutoff = 0.95         | 11     | 8.4    | 5.5    | 2.9    | 1      | -      |
| U3 - cutoff = 0.75         | 7.3    | 7.6    | 5.7    | 2.9    | 1.1    | -      |
| U3 - cutoff = 0.85         | 9.5    | 7.9    | 5.7    | 2.9    | 1.1    | -      |
| U3 - cutoff = 0.95         | 11.1   | 8.3    | 5.4    | 2.8    | 1      | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.8    | 3.3    | 1.6    | 1      | 0.3    | -      |
| U2 - cutoff = 0.85         | 2.7    | 3.4    | 1.6    | 0.9    | 0.3    | -      |
| U2 - cutoff = 0.95         | 0.2    | 3.1    | 1.1    | 0.8    | 0.2    | -      |
| U3 - cutoff = 0.75         | 1.8    | 3.3    | 1.6    | 1      | 0.3    | -      |
| U3 - cutoff = 0.85         | 2.7    | 3.4    | 1.6    | 0.9    | 0.3    | -      |
| U3 - cutoff = 0.95         | 0.2    | 3      | 1.1    | 0.8    | 0.2    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 925.9  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 997.5  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1291.6 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 925.1  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 997    |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1290.7 |
| Scenario 4                 |        |        |        |        |        |        |
| DLT rate                   | 0.05   | 0.10   | 0.18   | 0.30   | 0.40   |        |
| Efficacy rate              | 0.05   | 0.10   | 0.25   | 0.50   | 0.55   |        |
| Utility 2                  | 0      | 0      | 0.1    | 0.2    | 0.1    |        |
| Utility 3                  | 41     | 42     | 47.8   | 58     | 57     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 7.8    | 7.9    | 24.2   | 41.7   | 11     | 7.4    |
| U2 - cutoff = 0.85         | 8.4    | 10.6   | 24.7   | 41.7   | 10.9   | 3.7    |
| U2 - cutoff = 0.95         | 9.6    | 10     | 25.6   | 41.1   | 11.7   | 2      |
| U3 - cutoff = 0.75         | 11.1   | 10.1   | 21.9   | 39.2   | 10     | 7.7    |
| U3 - cutoff = 0.85         | 12.7   | 11.5   | 22.2   | 40     | 9.8    | 3.8    |
| U3 - cutoff = 0.95         | 13.8   | 11.6   | 22.3   | 41.4   | 8.9    | 2      |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.2    | 4.4    | 7.1    | 8.1    | 5      | -      |
| U2 - cutoff = 0.85         | 4.5    | 4.5    | 7.2    | 8      | 5      | -      |
| U2 - cutoff = 0.95         | 4.9    | 4.9    | 7.2    | 8.2    | 4.6    | -      |
| U3 - cutoff = 0.75         | 4.2    | 4.4    | 7.1    | 8.1    | 5      | -      |
| U3 - cutoff = 0.85         | 4.4    | 4.5    | 7.3    | 8.1    | 4.9    | -      |
| U3 - cutoff = 0.95         | 4.9    | 4.9    | 7.2    | 8.2    | 4.5    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.5    | 2.8    | 2.2    | 2.7    | 1.2    | -      |
| U2 - cutoff = 0.85         | 0.7    | 2.9    | 2.3    | 2.6    | 1.2    | -      |
| U2 - cutoff = 0.95         | 0.3    | 3      | 1.5    | 2.2    | 0.6    | -      |
| U3 - cutoff = 0.75         | 0.6    | 2.8    | 2.2    | 2.7    | 1.2    | -      |
| U3 - cutoff = 0.85         | 0.7    | 2.9    | 2.3    | 2.7    | 1.2    | -      |
| U3 - cutoff = 0.95         | 0.3    | 3      | 1.5    | 2.2    | 0.6    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1062.8 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1077.2 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1216.9 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1061.8 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1078   |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1217   |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 5                 |        |        |        |        |        |        |
| DLT rate                   | 0.07   | 0.12   | 0.17   | 0.22   | 0.30   |        |
| Efficacy rate              | 0.05   | 0.30   | 0.30   | 0.30   | 0.30   |        |
| Utility 2                  | 0      | 0.2    | 0.1    | 0.1    | 0      |        |
| Utility 3                  | 40.2   | 53.2   | 51.2   | 49.2   | 46     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 3      | 35.3   | 26.5   | 19.7   | 6.8    | 8.7    |
| U2 - cutoff = 0.85         | 3.5    | 35.7   | 27.5   | 19.1   | 7.1    | 7.1    |
| U2 - cutoff = 0.95         | 4.3    | 36.6   | 27.3   | 19.8   | 7.6    | 4.3    |
| U3 - cutoff = 0.75         | 5.8    | 39.8   | 25.2   | 16.2   | 4.3    | 8.7    |
| U3 - cutoff = 0.85         | 6.2    | 40.2   | 25.6   | 16.6   | 4.3    | 7.1    |
| U3 - cutoff = 0.95         | 7      | 43.3   | 23.7   | 17.3   | 4.3    | 4.4    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.2    | 6.6    | 6.7    | 6      | 4.6    | -      |
| U2 - cutoff = 0.85         | 4.6    | 6.8    | 6.7    | 5.9    | 4.5    | -      |
| U2 - cutoff = 0.95         | 5.1    | 6.9    | 6.8    | 6      | 4.4    | -      |
| U3 - cutoff = 0.75         | 4.2    | 6.6    | 6.6    | 6.1    | 4.5    | -      |
| U3 - cutoff = 0.85         | 4.6    | 6.7    | 6.7    | 6      | 4.5    | -      |
| U3 - cutoff = 0.95         | 5.1    | 7.2    | 6.8    | 5.9    | 4.2    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.5    | 3.5    | 2.1    | 2.3    | 1.6    | -      |
| U2 - cutoff = 0.85         | 0.7    | 3.6    | 2.1    | 2.2    | 1.6    | -      |
| U2 - cutoff = 0.95         | 0.2    | 3.4    | 1.7    | 2.1    | 1      | -      |
| U3 - cutoff = 0.75         | 0.6    | 3.5    | 2.1    | 2.3    | 1.6    | -      |
| U3 - cutoff = 0.85         | 0.7    | 3.6    | 2.1    | 2.2    | 1.6    | -      |
| U3 - cutoff = 0.95         | 0.2    | 3.4    | 1.7    | 2.2    | 0.9    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 990.8  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1007   |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1152.1 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 990.7  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1006.9 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1155.7 |
| Scenario 6                 |        |        |        |        |        |        |
| DLT rate                   | 0.29   | 0.34   | 0.38   | 0.43   | 0.52   |        |
| Efficacy rate              | 0.30   | 0.32   | 0.35   | 0.38   | 0.40   |        |
| Utility 2                  | 0      | 0      | 0      | -0.1   | -0.1   |        |
| Utility 3                  | 46.4   | 45.6   | 45.8   | 45.6   | 43.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 61.7   | 9.6    | 1.7    | 0.6    | 0      | 26.4   |
| U2 - cutoff = 0.85         | 66.2   | 9.7    | 1.5    | 0.6    | 0.1    | 21.9   |
| U2 - cutoff = 0.95         | 70.2   | 12.3   | 1.8    | 0.4    | 0.1    | 15.1   |
| U3 - cutoff = 0.75         | 63.1   | 8.9    | 1.5    | 0.3    | 0      | 26.2   |
| U3 - cutoff = 0.85         | 68.3   | 8.5    | 1.4    | 0.2    | 0.1    | 21.5   |
| U3 - cutoff = 0.95         | 70.4   | 12.7   | 1.2    | 0.3    | 0.1    | 15.3   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 19.1   | 3.6    | 1.6    | 0.6    | 0.1    | -      |
| U2 - cutoff = 0.85         | 20.1   | 3.6    | 1.7    | 0.5    | 0.2    | -      |
| U2 - cutoff = 0.95         | 20.8   | 4.1    | 1.7    | 0.5    | 0.2    | -      |
| U3 - cutoff = 0.75         | 19.2   | 3.6    | 1.7    | 0.6    | 0.2    | -      |
| U3 - cutoff = 0.85         | 20.1   | 3.6    | 1.7    | 0.5    | 0.2    | -      |
| U3 - cutoff = 0.95         | 20.9   | 4.1    | 1.6    | 0.5    | 0.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 5.3    | 1.5    | 0.5    | 0.2    | 0      | -      |
| U2 - cutoff = 0.85         | 5.6    | 1.5    | 0.5    | 0.2    | 0      | -      |
| U2 - cutoff = 0.95         | 0.4    | 1.5    | 0.4    | 0.2    | 0      | -      |
| U3 - cutoff = 0.75         | 5.3    | 1.5    | 0.5    | 0.2    | 0      | -      |
| U3 - cutoff = 0.85         | 5.6    | 1.5    | 0.5    | 0.2    | 0      | -      |
| U3 - cutoff = 0.95         | 0.4    | 1.5    | 0.4    | 0.2    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 973.1  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1011.5 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1374.3 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 975.7  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1011.8 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1373.5 |

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 7                 |        |        |        |        |        |        |
| DLT rate                   | 0.16   | 0.20   | 0.26   | 0.31   | 0.36   |        |
| Efficacy rate              | 0.20   | 0.50   | 0.50   | 0.50   | 0.50   |        |
| Utility 2                  | 0      | 0.3    | 0.2    | 0.2    | 0.1    |        |
| Utility 3                  | 45.6   | 62     | 59.6   | 57.6   | 55.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 23.4   | 38.1   | 17.8   | 7.6    | 1.6    | 11.5   |
| U2 - cutoff = 0.85         | 28.2   | 39.2   | 18.3   | 7.8    | 1.6    | 4.9    |
| U2 - cutoff = 0.95         | 26.8   | 46.1   | 15.9   | 7.4    | 1.1    | 2.6    |
| U3 - cutoff = 0.75         | 25.8   | 39.7   | 14.8   | 7.3    | 0.9    | 11.5   |
| U3 - cutoff = 0.85         | 31     | 41.7   | 14.7   | 6.6    | 1.1    | 4.9    |
| U3 - cutoff = 0.95         | 29.5   | 47.1   | 13.9   | 6.3    | 0.6    | 2.6    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 10.5   | 7.4    | 5.3    | 3.1    | 1.5    | -      |
| U2 - cutoff = 0.85         | 11.7   | 7.5    | 5.3    | 3.1    | 1.6    | -      |
| U2 - cutoff = 0.95         | 12.1   | 7.9    | 5.2    | 3      | 1.4    | -      |
| U3 - cutoff = 0.75         | 10.5   | 7.5    | 5.2    | 3      | 1.6    | -      |
| U3 - cutoff = 0.85         | 11.8   | 7.4    | 5.4    | 3      | 1.5    | -      |
| U3 - cutoff = 0.95         | 12.2   | 7.7    | 5.1    | 3.1    | 1.4    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 2.9    | 2.9    | 1.3    | 0.9    | 0.4    | -      |
| U2 - cutoff = 0.85         | 3.4    | 3      | 1.3    | 0.9    | 0.4    | -      |
| U2 - cutoff = 0.95         | 0.2    | 2.7    | 1      | 0.8    | 0.2    | -      |
| U3 - cutoff = 0.75         | 2.9    | 3      | 1.3    | 0.9    | 0.4    | -      |
| U3 - cutoff = 0.85         | 3.4    | 3      | 1.3    | 0.9    | 0.4    | -      |
| U3 - cutoff = 0.95         | 0.2    | 2.7    | 1      | 0.8    | 0.2    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1065   |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1108.5 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1358.1 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1065.2 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1108.7 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1355.8 |
| Scenario 8                 |        |        |        |        |        |        |
| DLT rate                   | 0.32   | 0.37   | 0.42   | 0.48   | 0.52   |        |
| Efficacy rate              | 0.10   | 0.15   | 0.20   | 0.30   | 0.40   |        |
| Utility 2                  | -0.2   | -0.2   | -0.2   | -0.2   | -0.1   |        |
| Utility 3                  | 33.2   | 34.2   | 35.2   | 38.8   | 43.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 13.4   | 3.7    | 0.8    | 0.1    | 0      | 82     |
| U2 - cutoff = 0.85         | 27.5   | 5.4    | 0.9    | 0.1    | 0      | 66.1   |
| U2 - cutoff = 0.95         | 42.6   | 7.3    | 1      | 0.2    | 0      | 48.9   |
| U3 - cutoff = 0.75         | 14.2   | 2.9    | 0.6    | 0.1    | 0      | 82.2   |
| U3 - cutoff = 0.85         | 28.8   | 4.6    | 0.7    | 0.1    | 0      | 65.8   |
| U3 - cutoff = 0.95         | 43.5   | 6.3    | 0.6    | 0.2    | 0      | 49.4   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 10.8   | 2.7    | 1      | 0.3    | 0.1    | -      |
| U2 - cutoff = 0.85         | 14.6   | 2.8    | 1.1    | 0.2    | 0.1    | -      |
| U2 - cutoff = 0.95         | 18.7   | 3.2    | 1.1    | 0.2    | 0.1    | -      |
| U3 - cutoff = 0.75         | 10.8   | 2.7    | 1.1    | 0.3    | 0.1    | -      |
| U3 - cutoff = 0.85         | 14.6   | 2.8    | 1.1    | 0.3    | 0.1    | -      |
| U3 - cutoff = 0.95         | 18.7   | 3.2    | 1      | 0.3    | 0.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 3      | 1.3    | 0.4    | 0.1    | 0      | -      |
| U2 - cutoff = 0.85         | 4.4    | 1.4    | 0.4    | 0.1    | 0      | -      |
| U2 - cutoff = 0.95         | 0.3    | 1.5    | 0.3    | 0.1    | 0      | -      |
| U3 - cutoff = 0.75         | 3      | 1.3    | 0.4    | 0.1    | 0      | -      |
| U3 - cutoff = 0.85         | 4.4    | 1.4    | 0.4    | 0.1    | 0      | -      |
| U3 - cutoff = 0.95         | 0.3    | 1.5    | 0.3    | 0.1    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 548.4  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 686.2  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1164.5 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 548    |
| U3 - cutoff = 0.85         |        |        |        |        |        | 686.3  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1161.5 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.



Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 9                 |        |        |        |        |        |        |
| DLT rate                   | 0.15   | 0.18   | 0.22   | 0.26   | 0.29   |        |
| Efficacy rate              | 0.05   | 0.08   | 0.20   | 0.25   | 0.50   |        |
| Utility 2                  | -0.1   | -0.1   | 0      | 0      | 0.2    |        |
| Utility 3                  | 37     | 37.6   | 43.2   | 44.6   | 58.4   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 8.3    | 8.9    | 14.3   | 11.6   | 15.8   | 41.1   |
| U2 - cutoff = 0.85         | 11.2   | 13.7   | 15.6   | 11.1   | 15.7   | 32.7   |
| U2 - cutoff = 0.95         | 18.9   | 17.3   | 17     | 11.8   | 14     | 21     |
| U3 - cutoff = 0.75         | 11.8   | 9      | 13.6   | 9.5    | 14.9   | 41.2   |
| U3 - cutoff = 0.85         | 15.3   | 12.7   | 14.4   | 10.3   | 14.4   | 32.9   |
| U3 - cutoff = 0.95         | 25.5   | 18.2   | 14.4   | 9.4    | 11.6   | 20.9   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 6.1    | 4.2    | 5.1    | 3.4    | 3.5    | -      |
| U2 - cutoff = 0.85         | 8      | 4.5    | 5.2    | 3.4    | 3.5    | -      |
| U2 - cutoff = 0.95         | 10     | 5.1    | 5.3    | 3.7    | 3.2    | -      |
| U3 - cutoff = 0.75         | 6.1    | 4.2    | 5.2    | 3.4    | 3.5    | -      |
| U3 - cutoff = 0.85         | 8      | 4.5    | 5.2    | 3.4    | 3.5    | -      |
| U3 - cutoff = 0.95         | 10.1   | 5.1    | 5.3    | 3.6    | 3.2    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.4    | 2.4    | 1.8    | 1.4    | 1      | -      |
| U2 - cutoff = 0.85         | 2.2    | 2.6    | 1.8    | 1.4    | 1.1    | -      |
| U2 - cutoff = 0.95         | 0.5    | 2.8    | 1.2    | 1.2    | 0.5    | -      |
| U3 - cutoff = 0.75         | 1.4    | 2.4    | 1.8    | 1.4    | 1      | -      |
| U3 - cutoff = 0.85         | 2.1    | 2.6    | 1.9    | 1.4    | 1.1    | -      |
| U3 - cutoff = 0.95         | 0.5    | 2.8    | 1.3    | 1.2    | 0.5    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 785    |
| U2 - cutoff = 0.85         |        |        |        |        |        | 859.5  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1157.6 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 783.5  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 859.4  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1156   |
|                            |        |        |        |        |        |        |
| Scenario 10                |        |        |        |        |        |        |
| DLT rate                   | 0.23   | 0.28   | 0.32   | 0.39   | 0.43   |        |
| Efficacy rate              | 0.05   | 0.40   | 0.42   | 0.45   | 0.50   |        |
| Utility 2                  | -0.2   | 0.1    | 0.1    | 0.1    | 0.1    |        |
| Utility 3                  | 33.8   | 52.8   | 52.4   | 51.4   | 52.8   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.6    | 21.4   | 9.1    | 1.7    | 0.7    | 62.5   |
| U2 - cutoff = 0.85         | 8.8    | 25     | 9.4    | 2.2    | 0.6    | 54     |
| U2 - cutoff = 0.95         | 13.2   | 31.9   | 8.3    | 2.4    | 0.9    | 43.3   |
| U3 - cutoff = 0.75         | 5      | 22.9   | 8      | 1.3    | 0.4    | 62.4   |
| U3 - cutoff = 0.85         | 9.4    | 26.1   | 8.2    | 1.4    | 0.6    | 54.3   |
| U3 - cutoff = 0.95         | 13.4   | 34     | 6.8    | 1.6    | 0.9    | 43.3   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 6.9    | 5.2    | 3.4    | 1.4    | 0.4    | -      |
| U2 - cutoff = 0.85         | 9.8    | 5.4    | 3.4    | 1.4    | 0.4    | -      |
| U2 - cutoff = 0.95         | 13.1   | 6.1    | 3.2    | 1.4    | 0.4    | -      |
| U3 - cutoff = 0.75         | 6.9    | 5.2    | 3.4    | 1.4    | 0.5    | -      |
| U3 - cutoff = 0.85         | 9.8    | 5.5    | 3.3    | 1.4    | 0.4    | -      |
| U3 - cutoff = 0.95         | 13.2   | 6.1    | 3.2    | 1.4    | 0.4    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.6    | 2.3    | 0.9    | 0.4    | 0.1    | -      |
| U2 - cutoff = 0.85         | 2.7    | 2.4    | 0.9    | 0.4    | 0.1    | -      |
| U2 - cutoff = 0.95         | 0.1    | 2.3    | 0.7    | 0.4    | 0.1    | -      |
| U3 - cutoff = 0.75         | 1.6    | 2.3    | 0.9    | 0.4    | 0.1    | -      |
| U3 - cutoff = 0.85         | 2.7    | 2.4    | 0.9    | 0.4    | 0.1    | -      |
| U3 - cutoff = 0.95         | 0.1    | 2.3    | 0.7    | 0.4    | 0.1    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 647.7  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 750.6  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1138.3 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 646.7  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 748.5  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1142.3 |

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 11                |        |        |        |        |        |        |
| DLT rate                   | 0.22   | 0.32   | 0.44   | 0.59   | 0.67   |        |
| Efficacy rate              | 0.20   | 0.20   | 0.20   | 0.20   | 0.20   |        |
| Utility 2                  | 0      | -0.1   | -0.2   | -0.4   | -0.5   |        |
| Utility 3                  | 43.2   | 39.2   | 34.4   | 28.4   | 25.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 55     | 15.5   | 1.2    | 0      | 0      | 28.3   |
| U2 - cutoff = 0.85         | 67.9   | 16.6   | 1.2    | 0      | 0      | 14.3   |
| U2 - cutoff = 0.95         | 66.8   | 23.7   | 1.5    | 0.2    | 0      | 7.7    |
| U3 - cutoff = 0.75         | 56.9   | 13.9   | 0.7    | 0      | 0      | 28.5   |
| U3 - cutoff = 0.85         | 69     | 15.8   | 0.9    | 0      | 0      | 14.3   |
| U3 - cutoff = 0.95         | 68.6   | 22.8   | 0.6    | 0.1    | 0      | 7.9    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 16.7   | 5.4    | 2.4    | 0.5    | 0      | -      |
| U2 - cutoff = 0.85         | 19.1   | 5.5    | 2.4    | 0.5    | 0      | -      |
| U2 - cutoff = 0.95         | 20     | 5.8    | 2.4    | 0.5    | 0      | -      |
| U3 - cutoff = 0.75         | 16.8   | 5.3    | 2.4    | 0.5    | 0      | -      |
| U3 - cutoff = 0.85         | 19.1   | 5.5    | 2.4    | 0.5    | 0      | -      |
| U3 - cutoff = 0.95         | 20     | 5.9    | 2.4    | 0.5    | 0      | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 5.1    | 2.4    | 0.8    | 0.2    | 0      | -      |
| U2 - cutoff = 0.85         | 6      | 2.4    | 0.8    | 0.2    | 0      | -      |
| U2 - cutoff = 0.95         | 0.6    | 2.3    | 0.6    | 0.2    | 0      | -      |
| U3 - cutoff = 0.75         | 5.1    | 2.4    | 0.8    | 0.2    | 0      | -      |
| U3 - cutoff = 0.85         | 6      | 2.5    | 0.8    | 0.2    | 0      | -      |
| U3 - cutoff = 0.95         | 0.6    | 2.3    | 0.6    | 0.2    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 907.5  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 994.2  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1384.9 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 905.7  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 994.8  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1388.1 |
|                            |        |        |        |        |        |        |
| Scenario 12                |        |        |        |        |        |        |
| DLT rate                   | 0.06   | 0.09   | 0.18   | 0.29   | 0.44   |        |
| Efficacy rate              | 0.05   | 0.10   | 0.12   | 0.40   | 0.60   |        |
| Utility 2                  | 0      | 0      | -0.1   | 0.1    | 0.2    |        |
| Utility 3                  | 40.6   | 42.4   | 40     | 52.4   | 58.4   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 12.2   | 13.8   | 11.8   | 36     | 12.8   | 13.4   |
| U2 - cutoff = 0.85         | 14.8   | 17.1   | 12.6   | 37.1   | 12.9   | 5.5    |
| U2 - cutoff = 0.95         | 17.4   | 16.4   | 12.6   | 36.1   | 14.1   | 3.3    |
| U3 - cutoff = 0.75         | 17.8   | 14.8   | 9.4    | 33.2   | 11.7   | 13.1   |
| U3 - cutoff = 0.85         | 21.1   | 17.6   | 10     | 33.4   | 12.1   | 5.8    |
| U3 - cutoff = 0.95         | 23.3   | 16.9   | 8.9    | 34.4   | 13.1   | 3.4    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.5    | 5.1    | 5.8    | 7.2    | 5.5    | -      |
| U2 - cutoff = 0.85         | 4.9    | 5.3    | 6      | 7.4    | 5.5    | -      |
| U2 - cutoff = 0.95         | 5.5    | 5.5    | 6      | 7.4    | 5.2    | -      |
| U3 - cutoff = 0.75         | 4.5    | 5.2    | 5.8    | 7.2    | 5.5    | -      |
| U3 - cutoff = 0.85         | 5      | 5.3    | 6      | 7.3    | 5.5    | -      |
| U3 - cutoff = 0.95         | 5.4    | 5.6    | 6      | 7.4    | 5.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.7    | 3.1    | 1.9    | 2.7    | 1.3    | -      |
| U2 - cutoff = 0.85         | 0.9    | 3.2    | 2      | 2.8    | 1.3    | -      |
| U2 - cutoff = 0.95         | 0.4    | 3.1    | 1.5    | 2.2    | 0.7    | -      |
| U3 - cutoff = 0.75         | 0.7    | 3.1    | 1.9    | 2.7    | 1.3    | -      |
| U3 - cutoff = 0.85         | 0.9    | 3.2    | 1.9    | 2.7    | 1.3    | -      |
| U3 - cutoff = 0.95         | 0.4    | 3.1    | 1.5    | 2.3    | 0.7    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1019.4 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1047.3 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1187.2 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1018.4 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1046.8 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1185.5 |

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 13                |        |        |        |        |        |        |
| DLT rate                   | 0.21   | 0.29   | 0.30   | 0.39   | 0.46   |        |
| Efficacy rate              | 0.05   | 0.30   | 0.32   | 0.35   | 0.40   |        |
| Utility 2                  | -0.2   | 0      | 0      | 0      | -0.1   |        |
| Utility 3                  | 34.6   | 46.4   | 47.2   | 45.4   | 45.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 5.8    | 22.5   | 9.4    | 1.9    | 0.5    | 59.9   |
| U2 - cutoff = 0.85         | 11.4   | 26.2   | 9.7    | 1.9    | 0.6    | 50.2   |
| U2 - cutoff = 0.95         | 14.9   | 33.6   | 10.2   | 2.3    | 0.6    | 38.5   |
| U3 - cutoff = 0.75         | 6.3    | 22.6   | 9      | 1.8    | 0.6    | 59.7   |
| U3 - cutoff = 0.85         | 12.1   | 26.6   | 9.2    | 1.4    | 0.5    | 50.2   |
| U3 - cutoff = 0.95         | 15.9   | 35.4   | 7.2    | 2.7    | 0.2    | 38.5   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 6.9    | 5.6    | 3.7    | 1.5    | 0.5    | -      |
| U2 - cutoff = 0.85         | 9.6    | 5.8    | 3.7    | 1.4    | 0.5    | -      |
| U2 - cutoff = 0.95         | 12.9   | 6.3    | 3.7    | 1.5    | 0.5    | -      |
| U3 - cutoff = 0.75         | 6.9    | 5.6    | 3.6    | 1.4    | 0.5    | -      |
| U3 - cutoff = 0.85         | 9.6    | 5.8    | 3.7    | 1.4    | 0.5    | -      |
| U3 - cutoff = 0.95         | 12.9   | 6.4    | 3.6    | 1.5    | 0.5    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.6    | 2.6    | 1.1    | 0.6    | 0.1    | -      |
| U2 - cutoff = 0.85         | 2.7    | 2.7    | 1.1    | 0.6    | 0.2    | -      |
| U2 - cutoff = 0.95         | 0.2    | 2.6    | 0.8    | 0.5    | 0.1    | -      |
| U3 - cutoff = 0.75         | 1.6    | 2.6    | 1.1    | 0.6    | 0.1    | -      |
| U3 - cutoff = 0.85         | 2.7    | 2.7    | 1.1    | 0.5    | 0.2    | -      |
| U3 - cutoff = 0.95         | 0.2    | 2.6    | 0.8    | 0.5    | 0.1    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 659.4  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 757.5  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1141.5 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 659.5  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 756.7  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1143.7 |
| Scenario 14                |        |        |        |        |        |        |
| DLT rate                   | 0.09   | 0.15   | 0.23   | 0.28   | 0.39   |        |
| Efficacy rate              | 0.05   | 0.08   | 0.12   | 0.30   | 0.40   |        |
| Utility 2                  | 0      | -0.1   | -0.1   | 0      | 0      |        |
| Utility 3                  | 39.4   | 38.8   | 38     | 46.8   | 48.4   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 12.5   | 12.6   | 12.1   | 22.5   | 9.9    | 30.4   |
| U2 - cutoff = 0.85         | 18     | 18.5   | 14     | 24.4   | 8.7    | 16.4   |
| U2 - cutoff = 0.95         | 24     | 19.5   | 15.1   | 24.4   | 8.9    | 8.1    |
| U3 - cutoff = 0.75         | 20.4   | 11.3   | 9.3    | 21.2   | 7.8    | 30     |
| U3 - cutoff = 0.85         | 24.8   | 16.8   | 11.7   | 21.8   | 7.8    | 17.1   |
| U3 - cutoff = 0.95         | 32.3   | 20.8   | 10.7   | 20.9   | 7      | 8.2    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 5.6    | 5.3    | 5.7    | 5.3    | 3.7    | -      |
| U2 - cutoff = 0.85         | 6.7    | 5.7    | 6.1    | 5.2    | 3.7    | -      |
| U2 - cutoff = 0.95         | 7.6    | 6.3    | 6      | 5.4    | 3.5    | -      |
| U3 - cutoff = 0.75         | 5.6    | 5.2    | 5.7    | 5.3    | 3.8    | -      |
| U3 - cutoff = 0.85         | 6.6    | 5.7    | 6      | 5.3    | 3.8    | -      |
| U3 - cutoff = 0.95         | 7.6    | 6.3    | 6      | 5.4    | 3.5    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.2    | 3      | 2      | 2.1    | 1.2    | -      |
| U2 - cutoff = 0.85         | 1.6    | 3.3    | 2.1    | 2      | 1.2    | -      |
| U2 - cutoff = 0.95         | 0.6    | 3.3    | 1.5    | 1.8    | 0.6    | -      |
| U3 - cutoff = 0.75         | 1.2    | 3      | 2      | 2      | 1.2    | -      |
| U3 - cutoff = 0.85         | 1.6    | 3.2    | 2.1    | 2.1    | 1.2    | -      |
| U3 - cutoff = 0.95         | 0.6    | 3.4    | 1.5    | 1.8    | 0.6    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 889    |
| U2 - cutoff = 0.85         |        |        |        |        |        | 947.5  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1159   |
| U3 - cutoff = 0.75         |        |        |        |        |        | 890.8  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 947.9  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1154.6 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 15                |        |        |        |        |        |        |
| DLT rate                   | 0.13   | 0.20   | 0.32   | 0.39   | 0.46   |        |
| Efficacy rate              | 0.10   | 0.10   | 0.10   | 0.10   | 0.10   |        |
| Utility 2                  | 0      | -0.1   | -0.2   | -0.3   | -0.4   |        |
| Utility 3                  | 40.8   | 38     | 33.2   | 30.4   | 27.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 30.1   | 18.9   | 7.2    | 2.8    | 0.1    | 40.9   |
| U2 - cutoff = 0.85         | 41.7   | 26.8   | 9.2    | 2.9    | 0.2    | 19.2   |
| U2 - cutoff = 0.95         | 49.4   | 30.7   | 8.3    | 2.9    | 0.3    | 8.4    |
| U3 - cutoff = 0.75         | 36.4   | 16.8   | 4.7    | 1.2    | 0.2    | 40.7   |
| U3 - cutoff = 0.85         | 50.2   | 24.2   | 5.1    | 1.1    | 0.2    | 19.2   |
| U3 - cutoff = 0.95         | 56.9   | 28.8   | 4.5    | 1.6    | 0.1    | 8      |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 9      | 6.9    | 5.2    | 2.5    | 0.8    | -      |
| U2 - cutoff = 0.85         | 11.3   | 7.4    | 5.4    | 2.5    | 0.8    | -      |
| U2 - cutoff = 0.95         | 12.7   | 7.8    | 5.3    | 2.4    | 0.8    | -      |
| U3 - cutoff = 0.75         | 9.1    | 6.9    | 5.2    | 2.6    | 0.8    | -      |
| U3 - cutoff = 0.85         | 11.2   | 7.5    | 5.4    | 2.4    | 0.9    | -      |
| U3 - cutoff = 0.95         | 12.6   | 8      | 5.2    | 2.4    | 0.8    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 2.5    | 3.5    | 1.8    | 1.1    | 0.3    | -      |
| U2 - cutoff = 0.85         | 3.5    | 3.7    | 1.9    | 1.1    | 0.3    | -      |
| U2 - cutoff = 0.95         | 1.1    | 3.6    | 1.4    | 1      | 0.2    | -      |
| U3 - cutoff = 0.75         | 2.5    | 3.5    | 1.8    | 1.1    | 0.3    | -      |
| U3 - cutoff = 0.85         | 3.4    | 3.8    | 1.9    | 1.1    | 0.4    | -      |
| U3 - cutoff = 0.95         | 1.1    | 3.6    | 1.4    | 1      | 0.2    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 832.7  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 923.1  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1194   |
| U3 - cutoff = 0.75         |        |        |        |        |        | 832.3  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 922.6  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1191.9 |
|                            |        |        |        |        |        |        |
| Scenario 16                |        |        |        |        |        |        |
| DLT rate                   | 0.02   | 0.06   | 0.09   | 0.14   | 0.26   |        |
| Efficacy rate              | 0.20   | 0.20   | 0.20   | 0.20   | 0.20   |        |
| Utility 2                  | 0.2    | 0.1    | 0.1    | 0.1    | -0.1   |        |
| Utility 3                  | 51.2   | 49.6   | 48.4   | 46.4   | 41.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 27.3   | 28.4   | 17.8   | 14.7   | 11.7   | 0.1    |
| U2 - cutoff = 0.85         | 27.8   | 28.1   | 17.3   | 15.2   | 11.5   | 0.1    |
| U2 - cutoff = 0.95         | 28.8   | 29.8   | 16.6   | 15     | 9.7    | 0.1    |
| U3 - cutoff = 0.75         | 34.2   | 32.7   | 15.3   | 10.9   | 6.8    | 0.1    |
| U3 - cutoff = 0.85         | 34.5   | 31.6   | 15     | 11.9   | 6.9    | 0.1    |
| U3 - cutoff = 0.95         | 35.4   | 34.1   | 14     | 11.5   | 4.9    | 0.1    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 5.3    | 5.5    | 5.3    | 6.9    | 6.9    | -      |
| U2 - cutoff = 0.85         | 5.8    | 5.9    | 5.5    | 5.7    | 7      | -      |
| U2 - cutoff = 0.95         | 5.8    | 6.1    | 5.6    | 6.1    | 6.5    | -      |
| U3 - cutoff = 0.75         | 5.3    | 5.5    | 5.3    | 6.9    | 7      | -      |
| U3 - cutoff = 0.85         | 5.8    | 5.9    | 5.5    | 5.8    | 7      | -      |
| U3 - cutoff = 0.95         | 5.7    | 6.1    | 5.6    | 6      | 6.6    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.9    | 3.1    | 1.8    | 3.1    | 2.6    | -      |
| U2 - cutoff = 0.85         | 1.2    | 3.3    | 1.9    | 2.6    | 2.7    | -      |
| U2 - cutoff = 0.95         | 1      | 3.3    | 1.8    | 2.7    | 1.5    | -      |
| U3 - cutoff = 0.75         | 1      | 3.2    | 1.8    | 3.1    | 2.6    | -      |
| U3 - cutoff = 0.85         | 1.1    | 3.3    | 1.9    | 2.7    | 2.7    | -      |
| U3 - cutoff = 0.95         | 1      | 3.3    | 1.8    | 2.6    | 1.5    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1008   |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1007.2 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1089.7 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1007.9 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1006.8 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1090.1 |



Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 17                |        |        |        |        |        |        |
| DLT rate                   | 0.11   | 0.23   | 0.39   | 0.58   | 0.76   |        |
| Efficacy rate              | 0.30   | 0.50   | 0.50   | 0.50   | 0.50   |        |
| Utility 2                  | 0.2    | 0.3    | 0.1    | -0.1   | -0.3   |        |
| Utility 3                  | 53.6   | 60.8   | 54.4   | 46.8   | 39.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 29.7   | 53.4   | 12.8   | 0.9    | 0      | 3.2    |
| U2 - cutoff = 0.85         | 31.5   | 53.9   | 13.1   | 1      | 0      | 0.5    |
| U2 - cutoff = 0.95         | 30.2   | 54.9   | 13.1   | 1.4    | 0      | 0.3    |
| U3 - cutoff = 0.75         | 32.6   | 52.9   | 10.7   | 0.7    | 0      | 3.1    |
| U3 - cutoff = 0.85         | 34.6   | 53.9   | 10.3   | 0.7    | 0      | 0.5    |
| U3 - cutoff = 0.95         | 33.7   | 54.1   | 11     | 0.9    | 0      | 0.3    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 10.7   | 10.5   | 6.5    | 1.5    | 0.2    | -      |
| U2 - cutoff = 0.85         | 11.1   | 10.5   | 6.5    | 1.5    | 0.2    | -      |
| U2 - cutoff = 0.95         | 11     | 10.9   | 6.3    | 1.5    | 0.2    | -      |
| U3 - cutoff = 0.75         | 10.7   | 10.4   | 6.6    | 1.6    | 0.2    | -      |
| U3 - cutoff = 0.85         | 11.1   | 10.5   | 6.5    | 1.6    | 0.2    | -      |
| U3 - cutoff = 0.95         | 11.2   | 10.8   | 6.3    | 1.5    | 0.2    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 2.7    | 3.6    | 1.5    | 0.5    | 0      | -      |
| U2 - cutoff = 0.85         | 2.9    | 3.6    | 1.5    | 0.5    | 0.1    | -      |
| U2 - cutoff = 0.95         | 0.4    | 2.9    | 1      | 0.5    | 0      | -      |
| U3 - cutoff = 0.75         | 2.7    | 3.6    | 1.5    | 0.5    | 0      | -      |
| U3 - cutoff = 0.85         | 2.9    | 3.6    | 1.5    | 0.5    | 0.1    | -      |
| U3 - cutoff = 0.95         | 0.4    | 2.9    | 1      | 0.5    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1167.6 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1183.5 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1384   |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1167.7 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1183.1 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1383.8 |
| Scenario 18                |        |        |        |        |        |        |
| DLT rate                   | 0.02   | 0.04   | 0.10   | 0.20   | 0.35   |        |
| Efficacy rate              | 0.05   | 0.10   | 0.15   | 0.20   | 0.50   |        |
| Utility 2                  | 0      | 0.1    | 0.1    | 0      | 0.1    |        |
| Utility 3                  | 42.2   | 44.4   | 45     | 44     | 56     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 9.7    | 12.3   | 10.9   | 16.8   | 48.1   | 2.2    |
| U2 - cutoff = 0.85         | 9.7    | 11.9   | 11.5   | 17.9   | 48     | 1      |
| U2 - cutoff = 0.95         | 10.3   | 11.9   | 11.6   | 18.3   | 47.1   | 0.8    |
| U3 - cutoff = 0.75         | 14.7   | 13.4   | 11     | 15.3   | 43.4   | 2.2    |
| U3 - cutoff = 0.85         | 14.5   | 14.3   | 11.2   | 16.1   | 43     | 0.9    |
| U3 - cutoff = 0.95         | 15.3   | 15.5   | 11.5   | 16.7   | 40.2   | 0.8    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 3.6    | 4      | 4.9    | 5.7    | 11.5   | -      |
| U2 - cutoff = 0.85         | 3.7    | 4      | 4.9    | 5.7    | 11.5   | -      |
| U2 - cutoff = 0.95         | 3.9    | 4.3    | 5.1    | 6.1    | 10.6   | -      |
| U3 - cutoff = 0.75         | 3.6    | 3.9    | 5      | 5.7    | 11.6   | -      |
| U3 - cutoff = 0.85         | 3.7    | 4      | 4.9    | 5.7    | 11.6   | -      |
| U3 - cutoff = 0.95         | 3.9    | 4.2    | 5.1    | 6.1    | 10.6   | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.3    | 2.8    | 1.6    | 2.5    | 3.2    | -      |
| U2 - cutoff = 0.85         | 0.3    | 2.8    | 1.6    | 2.6    | 3.2    | -      |
| U2 - cutoff = 0.95         | 0.3    | 2.9    | 1.4    | 2.4    | 1.4    | -      |
| U3 - cutoff = 0.75         | 0.3    | 2.7    | 1.7    | 2.6    | 3.2    | -      |
| U3 - cutoff = 0.85         | 0.3    | 2.8    | 1.6    | 2.6    | 3.2    | -      |
| U3 - cutoff = 0.95         | 0.2    | 2.9    | 1.5    | 2.4    | 1.4    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1062.3 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1065.9 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1184.8 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1062.6 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1066.6 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1185.9 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 19                |        |        |        |        |        |        |
| DLT rate                   | 0.11   | 0.26   | 0.39   | 0.60   | 0.77   |        |
| Efficacy rate              | 0.08   | 0.30   | 0.32   | 0.35   | 0.40   |        |
| Utility 2                  | 0      | 0      | -0.1   | -0.2   | -0.4   |        |
| Utility 3                  | 40.4   | 47.6   | 43.6   | 37     | 33.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 16.7   | 41.9   | 10.2   | 0.8    | 0      | 30.4   |
| U2 - cutoff = 0.85         | 22.2   | 48.8   | 11.2   | 0.5    | 0      | 17.3   |
| U2 - cutoff = 0.95         | 22.6   | 55     | 12.4   | 0.6    | 0      | 9.5    |
| U3 - cutoff = 0.75         | 18.8   | 41     | 8.9    | 0.6    | 0      | 30.7   |
| U3 - cutoff = 0.85         | 24.8   | 48.2   | 9.4    | 0.4    | 0      | 17.2   |
| U3 - cutoff = 0.95         | 25.8   | 53.6   | 10.4   | 0.5    | 0      | 9.8    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 7.3    | 10.2   | 5.9    | 1.3    | 0.2    | -      |
| U2 - cutoff = 0.85         | 9      | 10.6   | 6      | 1.3    | 0.2    | -      |
| U2 - cutoff = 0.95         | 10.5   | 10.9   | 5.8    | 1.3    | 0.1    | -      |
| U3 - cutoff = 0.75         | 7.2    | 10.1   | 6      | 1.3    | 0.2    | -      |
| U3 - cutoff = 0.85         | 9      | 10.6   | 6      | 1.3    | 0.2    | -      |
| U3 - cutoff = 0.95         | 10.6   | 10.9   | 5.7    | 1.4    | 0.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.7    | 4.4    | 1.7    | 0.5    | 0      | -      |
| U2 - cutoff = 0.85         | 2.5    | 4.6    | 1.7    | 0.5    | 0      | -      |
| U2 - cutoff = 0.95         | 0.4    | 3.6    | 1.1    | 0.5    | 0      | -      |
| U3 - cutoff = 0.75         | 1.7    | 4.4    | 1.7    | 0.5    | 0      | -      |
| U3 - cutoff = 0.85         | 2.5    | 4.6    | 1.7    | 0.5    | 0      | -      |
| U3 - cutoff = 0.95         | 0.4    | 3.6    | 1.1    | 0.5    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 905.6  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 974.7  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1271.1 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 905.6  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 976.4  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1269.6 |
|                            |        |        |        |        |        |        |
| Scenario 20                |        |        |        |        |        |        |
| DLT rate                   | 0.07   | 0.14   | 0.31   | 0.41   | 0.52   |        |
| Efficacy rate              | 0.05   | 0.25   | 0.25   | 0.25   | 0.25   |        |
| Utility 2                  | 0      | 0.1    | -0.1   | -0.2   | -0.3   |        |
| Utility 3                  | 40.2   | 49.4   | 42.6   | 38.6   | 34.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 9      | 45.3   | 25.4   | 6.5    | 0.4    | 13.4   |
| U2 - cutoff = 0.85         | 10.6   | 48.9   | 26.6   | 6.1    | 0.1    | 7.7    |
| U2 - cutoff = 0.95         | 12     | 49.7   | 27.4   | 6.1    | 0.3    | 4.5    |
| U3 - cutoff = 0.75         | 13     | 47.7   | 21.1   | 4.8    | 0.3    | 13.1   |
| U3 - cutoff = 0.85         | 14.8   | 49.7   | 22.7   | 4.7    | 0.4    | 7.7    |
| U3 - cutoff = 0.95         | 15.7   | 52.7   | 21.4   | 5.4    | 0.3    | 4.5    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.9    | 9.3    | 8.4    | 3.7    | 1.4    | -      |
| U2 - cutoff = 0.85         | 5.5    | 9.5    | 8.5    | 3.7    | 1.3    | -      |
| U2 - cutoff = 0.95         | 6.3    | 9.9    | 8.2    | 3.6    | 1.3    | -      |
| U3 - cutoff = 0.75         | 4.9    | 9.3    | 8.4    | 3.8    | 1.3    | -      |
| U3 - cutoff = 0.85         | 5.5    | 9.5    | 8.4    | 3.7    | 1.4    | -      |
| U3 - cutoff = 0.95         | 6.3    | 9.9    | 8.1    | 3.6    | 1.3    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.8    | 4.5    | 2.6    | 1.4    | 0.5    | -      |
| U2 - cutoff = 0.85         | 1.1    | 4.6    | 2.6    | 1.4    | 0.5    | -      |
| U2 - cutoff = 0.95         | 0.4    | 3.8    | 1.8    | 1.4    | 0.3    | -      |
| U3 - cutoff = 0.75         | 0.8    | 4.5    | 2.6    | 1.5    | 0.4    | -      |
| U3 - cutoff = 0.85         | 1.1    | 4.6    | 2.6    | 1.4    | 0.5    | -      |
| U3 - cutoff = 0.95         | 0.4    | 3.8    | 1.8    | 1.4    | 0.3    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 979.1  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1004.6 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1196.5 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 981    |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1004.9 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1196.2 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 21                |        |        |        |        |        |        |
| DLT rate                   | 0.37   | 0.56   | 0.69   | 0.83   | 0.88   |        |
| Efficacy rate              | 0.50   | 0.50   | 0.50   | 0.50   | 0.50   |        |
| Utility 2                  | 0.1    | -0.1   | -0.2   | -0.3   | -0.4   |        |
| Utility 3                  | 55.2   | 47.6   | 42.4   | 36.8   | 34.8   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 47.6   | 0.4    | 0      | 0      | 0      | 52     |
| U2 - cutoff = 0.85         | 47.6   | 0.4    | 0      | 0      | 0      | 52     |
| U2 - cutoff = 0.95         | 57.8   | 1.7    | 0      | 0      | 0      | 40.5   |
| U3 - cutoff = 0.75         | 47.3   | 0.7    | 0      | 0      | 0      | 52     |
| U3 - cutoff = 0.85         | 47.3   | 0.4    | 0      | 0      | 0      | 52.3   |
| U3 - cutoff = 0.95         | 57.5   | 1.5    | 0      | 0      | 0      | 41     |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 19.7   | 1.5    | 0.3    | 0      | 0      | -      |
| U2 - cutoff = 0.85         | 19.7   | 1.6    | 0.3    | 0      | 0      | -      |
| U2 - cutoff = 0.95         | 21.4   | 1.7    | 0.3    | 0      | 0      | -      |
| U3 - cutoff = 0.75         | 19.7   | 1.6    | 0.3    | 0      | 0      | -      |
| U3 - cutoff = 0.85         | 19.7   | 1.6    | 0.3    | 0      | 0      | -      |
| U3 - cutoff = 0.95         | 21.3   | 1.7    | 0.3    | 0      | 0      | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4      | 0.5    | 0.1    | 0      | 0      | -      |
| U2 - cutoff = 0.85         | 4      | 0.5    | 0.1    | 0      | 0      | -      |
| U2 - cutoff = 0.95         | 0.1    | 0.6    | 0.1    | 0      | 0      | -      |
| U3 - cutoff = 0.75         | 4      | 0.5    | 0.1    | 0      | 0      | -      |
| U3 - cutoff = 0.85         | 4      | 0.5    | 0.1    | 0      | 0      | -      |
| U3 - cutoff = 0.95         | 0.1    | 0.6    | 0.1    | 0      | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 933.1  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 933.8  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1249.5 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 934.7  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 934.4  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1247.5 |
| Scenario 22                |        |        |        |        |        |        |
| DLT rate                   | 0.07   | 0.27   | 0.50   | 0.56   | 0.78   |        |
| Efficacy rate              | 0.08   | 0.25   | 0.25   | 0.50   | 0.50   |        |
| Utility 2                  | 0      | 0      | -0.2   | -0.1   | -0.3   |        |
| Utility 3                  | 42     | 44.2   | 35     | 47.6   | 38.8   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 20.1   | 43.7   | 7.4    | 0.4    | 0      | 28.4   |
| U2 - cutoff = 0.85         | 26.4   | 53     | 8.9    | 0.5    | 0      | 11.2   |
| U2 - cutoff = 0.95         | 25.3   | 59.8   | 9.8    | 0.2    | 0      | 4.9    |
| U3 - cutoff = 0.75         | 23.4   | 42.7   | 5.8    | 0.4    | 0      | 27.7   |
| U3 - cutoff = 0.85         | 29.6   | 52.3   | 6.9    | 0.4    | 0      | 10.8   |
| U3 - cutoff = 0.95         | 30.8   | 56.2   | 7.7    | 0.3    | 0      | 5      |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 7.5    | 12.2   | 5.2    | 1      | 0.1    | -      |
| U2 - cutoff = 0.85         | 8.8    | 13     | 5.2    | 1      | 0.1    | -      |
| U2 - cutoff = 0.95         | 10     | 12.9   | 5.3    | 1      | 0.1    | -      |
| U3 - cutoff = 0.75         | 7.4    | 12.2   | 5.2    | 1      | 0.1    | -      |
| U3 - cutoff = 0.85         | 8.7    | 13     | 5.3    | 1      | 0.1    | -      |
| U3 - cutoff = 0.95         | 10     | 12.9   | 5.3    | 1      | 0.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.8    | 5.3    | 1.5    | 0.4    | 0      | -      |
| U2 - cutoff = 0.85         | 2.3    | 5.6    | 1.6    | 0.4    | 0      | -      |
| U2 - cutoff = 0.95         | 0.7    | 4      | 1.1    | 0.4    | 0      | -      |
| U3 - cutoff = 0.75         | 1.7    | 5.3    | 1.5    | 0.4    | 0      | -      |
| U3 - cutoff = 0.85         | 2.3    | 5.6    | 1.5    | 0.4    | 0      | -      |
| U3 - cutoff = 0.95         | 0.7    | 4      | 1.1    | 0.4    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 927.7  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 994.8  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1274.2 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 927.6  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 995.4  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1272.3 |

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 23                |        |        |        |        |        |        |
| DLT rate                   | 0.32   | 0.55   | 0.80   | 0.94   | 0.97   |        |
| Efficacy rate              | 0.05   | 0.05   | 0.05   | 0.05   | 0.05   |        |
| Utility 2                  | -0.3   | -0.5   | -0.8   | -0.9   | -0.9   |        |
| Utility 3                  | 30.2   | 21     | 11     | 5.4    | 4.2    |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.4    | 0.5    | 0      | 0      | 0      | 98.1   |
| U2 - cutoff = 0.85         | 6.3    | 1.6    | 0      | 0      | 0      | 92.1   |
| U2 - cutoff = 0.95         | 16.2   | 5.7    | 0      | 0      | 0      | 78     |
| U3 - cutoff = 0.75         | 1.4    | 0.5    | 0      | 0      | 0      | 98.1   |
| U3 - cutoff = 0.85         | 6.3    | 1.6    | 0      | 0      | 0      | 92.1   |
| U3 - cutoff = 0.95         | 16.9   | 5.6    | 0      | 0      | 0      | 77.4   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 8.1    | 1.6    | 0.3    | 0      | 0      | -      |
| U2 - cutoff = 0.85         | 11.8   | 1.7    | 0.2    | 0      | 0      | -      |
| U2 - cutoff = 0.95         | 17.3   | 2.1    | 0.3    | 0      | 0      | -      |
| U3 - cutoff = 0.75         | 8.1    | 1.6    | 0.2    | 0      | 0      | -      |
| U3 - cutoff = 0.85         | 11.8   | 1.7    | 0.2    | 0      | 0      | -      |
| U3 - cutoff = 0.95         | 17.3   | 2.1    | 0.3    | 0      | 0      | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 2.1    | 1      | 0.1    | 0      | 0      | -      |
| U2 - cutoff = 0.85         | 3.4    | 1      | 0.1    | 0      | 0      | -      |
| U2 - cutoff = 0.95         | 0.2    | 1.3    | 0.1    | 0      | 0      | -      |
| U3 - cutoff = 0.75         | 2.1    | 1      | 0.1    | 0      | 0      | -      |
| U3 - cutoff = 0.85         | 3.4    | 1      | 0.1    | 0      | 0      | -      |
| U3 - cutoff = 0.95         | 0.2    | 1.3    | 0.1    | 0      | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 365.6  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 498.7  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 995    |
| U3 - cutoff = 0.75         |        |        |        |        |        | 365.1  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 498.3  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 994.8  |
|                            |        |        |        |        |        |        |
| Scenario 24                |        |        |        |        |        |        |
| DLT rate                   | 0.01   | 0.02   | 0.05   | 0.10   | 0.25   |        |
| Efficacy rate              | 0.05   | 0.08   | 0.20   | 0.50   | 0.50   |        |
| Utility 2                  | 0      | 0.1    | 0.1    | 0.4    | 0.2    |        |
| Utility 3                  | 42.6   | 44     | 50     | 66     | 60     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.6    | 1.5    | 3.7    | 47.8   | 46.1   | 0.3    |
| U2 - cutoff = 0.85         | 0.6    | 1.7    | 4.3    | 47.6   | 45.7   | 0.1    |
| U2 - cutoff = 0.95         | 1.5    | 1.7    | 5.4    | 53.3   | 38.1   | 0      |
| U3 - cutoff = 0.75         | 1.7    | 2.3    | 4.8    | 49.9   | 41     | 0.3    |
| U3 - cutoff = 0.85         | 1.7    | 2.4    | 5      | 50.5   | 40.3   | 0.1    |
| U3 - cutoff = 0.95         | 2.3    | 3.2    | 7.8    | 54     | 32.8   | 0      |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 3.2    | 3.1    | 3.6    | 8.4    | 11.7   | -      |
| U2 - cutoff = 0.85         | 3.2    | 3.1    | 3.6    | 8.3    | 11.8   | -      |
| U2 - cutoff = 0.95         | 3.2    | 3.2    | 3.8    | 9.1    | 10.6   | -      |
| U3 - cutoff = 0.75         | 3.2    | 3.1    | 3.6    | 8.4    | 11.6   | -      |
| U3 - cutoff = 0.85         | 3.2    | 3.1    | 3.6    | 8.3    | 11.7   | -      |
| U3 - cutoff = 0.95         | 3.3    | 3.2    | 3.9    | 9.1    | 10.6   | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.1    | 2.4    | 1.1    | 3.2    | 3.2    | -      |
| U2 - cutoff = 0.85         | 0.1    | 2.4    | 1.1    | 3.2    | 3.2    | -      |
| U2 - cutoff = 0.95         | 0.1    | 2.5    | 1.2    | 3.2    | 1.1    | -      |
| U3 - cutoff = 0.75         | 0.1    | 2.4    | 1.1    | 3.3    | 3.2    | -      |
| U3 - cutoff = 0.85         | 0.1    | 2.4    | 1.1    | 3.2    | 3.2    | -      |
| U3 - cutoff = 0.95         | 0.1    | 2.5    | 1.2    | 3.2    | 1.1    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1101.2 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1101.3 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1207.7 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1100.5 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1101.1 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1207.8 |



Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 25                |        |        |        |        |        |        |
| DLT rate                   | 0.09   | 0.18   | 0.37   | 0.63   | 0.77   |        |
| Efficacy rate              | 0.20   | 0.25   | 0.40   | 0.45   | 0.50   |        |
| Utility 2                  | 0.1    | 0.1    | 0      | -0.2   | -0.3   |        |
| Utility 3                  | 48.4   | 47.8   | 49.2   | 41.8   | 39.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 33.4   | 32.4   | 24.5   | 3.6    | 0      | 6.1    |
| U2 - cutoff = 0.85         | 36.2   | 33.5   | 24.9   | 3.3    | 0      | 2.1    |
| U2 - cutoff = 0.95         | 33.3   | 36.5   | 25.8   | 3.8    | 0      | 0.6    |
| U3 - cutoff = 0.75         | 37.2   | 32.3   | 21.6   | 2.6    | 0      | 6.3    |
| U3 - cutoff = 0.85         | 40.4   | 34.2   | 20.7   | 2.7    | 0      | 2      |
| U3 - cutoff = 0.95         | 38.8   | 35.1   | 22     | 3.4    | 0      | 0.7    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 9.6    | 8.7    | 8.5    | 2      | 0.2    | -      |
| U2 - cutoff = 0.85         | 10.1   | 8.7    | 8.6    | 1.9    | 0.3    | -      |
| U2 - cutoff = 0.95         | 10.3   | 9.3    | 8.2    | 1.9    | 0.2    | -      |
| U3 - cutoff = 0.75         | 9.7    | 8.5    | 8.6    | 2      | 0.2    | -      |
| U3 - cutoff = 0.85         | 10.1   | 8.8    | 8.6    | 1.9    | 0.2    | -      |
| U3 - cutoff = 0.95         | 10.1   | 9.4    | 8.1    | 2      | 0.3    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 2.7    | 3.8    | 2.3    | 0.7    | 0.1    | -      |
| U2 - cutoff = 0.85         | 2.9    | 3.8    | 2.4    | 0.7    | 0.1    | -      |
| U2 - cutoff = 0.95         | 0.8    | 3.3    | 1.5    | 0.7    | 0      | -      |
| U3 - cutoff = 0.75         | 2.7    | 3.7    | 2.4    | 0.7    | 0.1    | -      |
| U3 - cutoff = 0.85         | 2.8    | 3.8    | 2.4    | 0.7    | 0.1    | -      |
| U3 - cutoff = 0.95         | 0.7    | 3.3    | 1.4    | 0.7    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1076.1 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1095.2 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1304.4 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1075.4 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1094.7 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1303.5 |
| Scenario 26                |        |        |        |        |        |        |
| DLT rate                   | 0.15   | 0.22   | 0.30   | 0.62   | 0.80   |        |
| Efficacy rate              | 0.10   | 0.20   | 0.35   | 0.45   | 0.55   |        |
| Utility 2                  | -0.1   | 0      | 0.1    | -0.2   | -0.2   |        |
| Utility 3                  | 40     | 43.2   | 49     | 42.2   | 41     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 19.1   | 24.7   | 19.8   | 2.6    | 0      | 33.8   |
| U2 - cutoff = 0.85         | 27.3   | 29.6   | 20.9   | 2.5    | 0      | 19.7   |
| U2 - cutoff = 0.95         | 30.6   | 34.3   | 19.5   | 4      | 0      | 11.6   |
| U3 - cutoff = 0.75         | 21.7   | 24.8   | 17.5   | 2.3    | 0      | 33.7   |
| U3 - cutoff = 0.85         | 30.3   | 29.3   | 18.2   | 2      | 0      | 20.2   |
| U3 - cutoff = 0.95         | 34     | 34.6   | 15.8   | 3.5    | 0      | 12.1   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 8.4    | 6.9    | 6.8    | 1.8    | 0.2    | -      |
| U2 - cutoff = 0.85         | 10.7   | 7.3    | 6.9    | 1.8    | 0.2    | -      |
| U2 - cutoff = 0.95         | 12.5   | 7.6    | 6.5    | 1.8    | 0.2    | -      |
| U3 - cutoff = 0.75         | 8.5    | 6.9    | 6.7    | 1.9    | 0.2    | -      |
| U3 - cutoff = 0.85         | 10.8   | 7.3    | 6.8    | 1.8    | 0.3    | -      |
| U3 - cutoff = 0.95         | 12.5   | 7.7    | 6.4    | 1.9    | 0.2    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 2.3    | 3.3    | 2      | 0.7    | 0.1    | -      |
| U2 - cutoff = 0.85         | 3.2    | 3.4    | 2      | 0.7    | 0      | -      |
| U2 - cutoff = 0.95         | 0.5    | 3.1    | 1.1    | 0.7    | 0      | -      |
| U3 - cutoff = 0.75         | 2.3    | 3.3    | 2      | 0.7    | 0.1    | -      |
| U3 - cutoff = 0.85         | 3.2    | 3.4    | 2      | 0.7    | 0.1    | -      |
| U3 - cutoff = 0.95         | 0.5    | 3.1    | 1.1    | 0.7    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 879.7  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 964.2  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1282   |
| U3 - cutoff = 0.75         |        |        |        |        |        | 878.7  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 963.5  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1283.4 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 27                |        |        |        |        |        |        |
| DLT rate                   | 0.11   | 0.17   | 0.30   | 0.50   | 0.62   |        |
| Efficacy rate              | 0.10   | 0.20   | 0.40   | 0.42   | 0.45   |        |
| Utility 2                  | 0      | 0      | 0.1    | -0.1   | -0.2   |        |
| Utility 3                  | 41.6   | 45.2   | 52     | 45.2   | 42.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 15     | 24.9   | 35.4   | 4.4    | 0      | 20.3   |
| U2 - cutoff = 0.85         | 19.1   | 27.6   | 37.8   | 3.5    | 0      | 12     |
| U2 - cutoff = 0.95         | 21.7   | 30.6   | 35.4   | 6.5    | 0.1    | 5.7    |
| U3 - cutoff = 0.75         | 17.7   | 25.7   | 32.3   | 3.8    | 0      | 20.5   |
| U3 - cutoff = 0.85         | 23     | 28.7   | 33     | 3.4    | 0      | 11.9   |
| U3 - cutoff = 0.95         | 26.2   | 32     | 30.4   | 5.6    | 0      | 5.8    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 6.8    | 6.7    | 9.1    | 3.1    | 0.7    | -      |
| U2 - cutoff = 0.85         | 8.2    | 7.1    | 9      | 3.1    | 0.8    | -      |
| U2 - cutoff = 0.95         | 9      | 7.7    | 8.8    | 3      | 0.8    | -      |
| U3 - cutoff = 0.75         | 6.8    | 6.8    | 9      | 3.1    | 0.8    | -      |
| U3 - cutoff = 0.85         | 8.2    | 7.1    | 9      | 3.1    | 0.8    | -      |
| U3 - cutoff = 0.95         | 9.2    | 7.6    | 8.7    | 3.1    | 0.7    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.7    | 3.3    | 2.6    | 1      | 0.2    | -      |
| U2 - cutoff = 0.85         | 2.2    | 3.4    | 2.6    | 1      | 0.2    | -      |
| U2 - cutoff = 0.95         | 0.5    | 3.2    | 1.6    | 0.9    | 0.2    | -      |
| U3 - cutoff = 0.75         | 1.7    | 3.3    | 2.6    | 1      | 0.2    | -      |
| U3 - cutoff = 0.85         | 2.2    | 3.4    | 2.6    | 1      | 0.2    | -      |
| U3 - cutoff = 0.95         | 0.5    | 3.1    | 1.6    | 0.9    | 0.1    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 975.9  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1030   |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1272.6 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 975.8  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1029.8 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1270.6 |
| Scenario 28                |        |        |        |        |        |        |
| DLT rate                   | 0.20   | 0.30   | 0.50   | 0.60   | 0.77   |        |
| Efficacy rate              | 0.05   | 0.30   | 0.32   | 0.35   | 0.40   |        |
| Utility 2                  | -0.1   | 0      | -0.2   | -0.2   | -0.4   |        |
| Utility 3                  | 35     | 46     | 39.2   | 37     | 33.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 11.5   | 22.9   | 1      | 0      | 0      | 64.6   |
| U2 - cutoff = 0.85         | 18.2   | 28     | 1.2    | 0      | 0      | 52.6   |
| U2 - cutoff = 0.95         | 21     | 38.3   | 2.9    | 0.1    | 0      | 37.7   |
| U3 - cutoff = 0.75         | 11.9   | 22.9   | 0.9    | 0      | 0      | 64.3   |
| U3 - cutoff = 0.85         | 19     | 27.7   | 1      | 0      | 0      | 52.3   |
| U3 - cutoff = 0.95         | 22.3   | 37.3   | 2.4    | 0      | 0      | 38.1   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 7.4    | 7.2    | 2.9    | 0.5    | 0      | -      |
| U2 - cutoff = 0.85         | 10.2   | 7.4    | 3      | 0.4    | 0      | -      |
| U2 - cutoff = 0.95         | 13.7   | 7.9    | 3      | 0.4    | 0      | -      |
| U3 - cutoff = 0.75         | 7.4    | 7.1    | 2.9    | 0.4    | 0      | -      |
| U3 - cutoff = 0.85         | 10.2   | 7.5    | 2.9    | 0.4    | 0      | -      |
| U3 - cutoff = 0.95         | 13.8   | 7.9    | 2.9    | 0.4    | 0      | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.8    | 3.1    | 0.9    | 0.2    | 0      | -      |
| U2 - cutoff = 0.85         | 2.9    | 3.2    | 0.9    | 0.2    | 0      | -      |
| U2 - cutoff = 0.95         | 0.3    | 2.7    | 0.6    | 0.2    | 0      | -      |
| U3 - cutoff = 0.75         | 1.8    | 3.1    | 0.9    | 0.2    | 0      | -      |
| U3 - cutoff = 0.85         | 2.9    | 3.2    | 0.9    | 0.2    | 0      | -      |
| U3 - cutoff = 0.95         | 0.3    | 2.7    | 0.6    | 0.2    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 653.6  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 760.5  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1168.8 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 652.8  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 762.4  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1167.3 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 29                |        |        |        |        |        |        |
| DLT rate                   | 0.19   | 0.30   | 0.56   | 0.74   | 0.86   |        |
| Efficacy rate              | 0.40   | 0.40   | 0.40   | 0.40   | 0.40   |        |
| Utility 2                  | 0.2    | 0.1    | -0.2   | -0.3   | -0.5   |        |
| Utility 3                  | 56.4   | 52     | 41.6   | 34.4   | 29.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 73.4   | 20.3   | 1.5    | 0      | 0      | 4.8    |
| U2 - cutoff = 0.85         | 74.2   | 20.8   | 1.7    | 0      | 0      | 3.3    |
| U2 - cutoff = 0.95         | 68.1   | 26.3   | 2.3    | 0      | 0      | 3.3    |
| U3 - cutoff = 0.75         | 75.4   | 18.8   | 1.2    | 0      | 0      | 4.6    |
| U3 - cutoff = 0.85         | 76.7   | 18.8   | 1.1    | 0      | 0      | 3.4    |
| U3 - cutoff = 0.95         | 69.5   | 25.4   | 1.7    | 0      | 0      | 3.4    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 19.2   | 6.9    | 2.6    | 0.3    | 0      | -      |
| U2 - cutoff = 0.85         | 19.5   | 7      | 2.6    | 0.3    | 0      | -      |
| U2 - cutoff = 0.95         | 19.1   | 7.4    | 2.7    | 0.3    | 0      | -      |
| U3 - cutoff = 0.75         | 19.2   | 7      | 2.6    | 0.3    | 0      | -      |
| U3 - cutoff = 0.85         | 19.4   | 7      | 2.7    | 0.3    | 0      | -      |
| U3 - cutoff = 0.95         | 19.1   | 7.4    | 2.6    | 0.3    | 0      | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 5      | 2.3    | 0.7    | 0.1    | 0      | -      |
| U2 - cutoff = 0.85         | 5.1    | 2.3    | 0.7    | 0.1    | 0      | -      |
| U2 - cutoff = 0.95         | 0.6    | 2.1    | 0.6    | 0.1    | 0      | -      |
| U3 - cutoff = 0.75         | 5      | 2.3    | 0.7    | 0.1    | 0      | -      |
| U3 - cutoff = 0.85         | 5.1    | 2.3    | 0.8    | 0.1    | 0      | -      |
| U3 - cutoff = 0.95         | 0.6    | 2.1    | 0.6    | 0.1    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1156.2 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1165.7 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1439   |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1156.9 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1164.9 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1439.9 |
| Scenario 30                |        |        |        |        |        |        |
| DLT rate                   | 0.05   | 0.08   | 0.17   | 0.27   | 0.54   |        |
| Efficacy rate              | 0.12   | 0.15   | 0.20   | 0.40   | 0.50   |        |
| Utility 2                  | 0.1    | 0.1    | 0      | 0.1    | 0      |        |
| Utility 3                  | 45.2   | 45.8   | 45.2   | 53.2   | 48.4   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 17.9   | 16.9   | 16.9   | 36.7   | 7.1    | 4.5    |
| U2 - cutoff = 0.85         | 17.9   | 18.1   | 18.6   | 36.6   | 6.9    | 1.9    |
| U2 - cutoff = 0.95         | 20.7   | 19.7   | 15.8   | 34.5   | 8.6    | 0.8    |
| U3 - cutoff = 0.75         | 23.4   | 18.8   | 16.1   | 31.9   | 5.5    | 4.3    |
| U3 - cutoff = 0.85         | 23.9   | 20.4   | 14.1   | 33.9   | 5.7    | 2      |
| U3 - cutoff = 0.95         | 26.4   | 20.8   | 14.6   | 31     | 6.4    | 0.8    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 5      | 5      | 6.3    | 8.1    | 4.9    | -      |
| U2 - cutoff = 0.85         | 5.3    | 4.9    | 6.4    | 8.2    | 5      | -      |
| U2 - cutoff = 0.95         | 5.5    | 5.2    | 6.6    | 8      | 4.6    | -      |
| U3 - cutoff = 0.75         | 4.9    | 4.9    | 6.4    | 8.2    | 4.9    | -      |
| U3 - cutoff = 0.85         | 5.3    | 5      | 6.3    | 8.1    | 5      | -      |
| U3 - cutoff = 0.95         | 5.5    | 5.3    | 6.7    | 7.9    | 4.6    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.9    | 2.9    | 2      | 2.8    | 1.3    | -      |
| U2 - cutoff = 0.85         | 1      | 2.9    | 2.1    | 2.8    | 1.3    | -      |
| U2 - cutoff = 0.95         | 0.5    | 3      | 1.7    | 2.3    | 0.8    | -      |
| U3 - cutoff = 0.75         | 0.9    | 2.9    | 2.1    | 2.8    | 1.3    | -      |
| U3 - cutoff = 0.85         | 1      | 2.9    | 2      | 2.8    | 1.3    | -      |
| U3 - cutoff = 0.95         | 0.5    | 3      | 1.7    | 2.3    | 0.8    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1068.5 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1081.9 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1196.9 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1070   |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1080.9 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1191.7 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 31                |        |        |        |        |        |        |
| DLT rate                   | 0.04   | 0.06   | 0.10   | 0.20   | 0.60   |        |
| Efficacy rate              | 0.20   | 0.50   | 0.50   | 0.50   | 0.50   |        |
| Utility 2                  | 0.2    | 0.4    | 0.4    | 0.3    | -0.1   |        |
| Utility 3                  | 50.4   | 67.6   | 66     | 62     | 46     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 5.1    | 36.3   | 29.6   | 23.1   | 5.6    | 0.3    |
| U2 - cutoff = 0.85         | 5.5    | 36.2   | 30.4   | 22.2   | 5.5    | 0.2    |
| U2 - cutoff = 0.95         | 5.5    | 36.5   | 29.3   | 23.1   | 5.5    | 0.1    |
| U3 - cutoff = 0.75         | 7.9    | 41.4   | 28.9   | 18     | 3.5    | 0.3    |
| U3 - cutoff = 0.85         | 7.5    | 41.6   | 28.5   | 18.2   | 4      | 0.2    |
| U3 - cutoff = 0.95         | 7.7    | 45.1   | 24.2   | 19     | 4      | 0.1    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.1    | 5.8    | 6.5    | 8      | 5.5    | -      |
| U2 - cutoff = 0.85         | 4.2    | 5.8    | 6.6    | 8      | 5.5    | -      |
| U2 - cutoff = 0.95         | 4.1    | 6.1    | 6.8    | 7.9    | 5.1    | -      |
| U3 - cutoff = 0.75         | 4.1    | 5.8    | 6.5    | 8      | 5.5    | -      |
| U3 - cutoff = 0.85         | 4.1    | 5.8    | 6.6    | 8      | 5.5    | -      |
| U3 - cutoff = 0.95         | 4.1    | 6.2    | 6.6    | 7.9    | 5.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.5    | 2.9    | 1.7    | 2.1    | 1.4    | -      |
| U2 - cutoff = 0.85         | 0.5    | 2.9    | 1.7    | 2.1    | 1.4    | -      |
| U2 - cutoff = 0.95         | 0.1    | 2.8    | 1.5    | 1.7    | 1.1    | -      |
| U3 - cutoff = 0.75         | 0.5    | 2.9    | 1.7    | 2.1    | 1.4    | -      |
| U3 - cutoff = 0.85         | 0.5    | 2.9    | 1.7    | 2      | 1.4    | -      |
| U3 - cutoff = 0.95         | 0.1    | 2.9    | 1.4    | 1.8    | 1.1    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1182.1 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1182.8 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1252.3 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1181.7 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1183.7 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1251.6 |
|                            |        |        |        |        |        |        |
| Scenario 32                |        |        |        |        |        |        |
| DLT rate                   | 0.02   | 0.05   | 0.11   | 0.30   | 0.36   |        |
| Efficacy rate              | 0.08   | 0.15   | 0.30   | 0.30   | 0.30   |        |
| Utility 2                  | 0.1    | 0.1    | 0.2    | 0      | -0.1   |        |
| Utility 3                  | 44     | 47     | 53.6   | 46     | 43.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 9.4    | 14.7   | 34.4   | 29.6   | 10.8   | 1.1    |
| U2 - cutoff = 0.85         | 9.5    | 14.2   | 35.9   | 29.5   | 10.2   | 0.7    |
| U2 - cutoff = 0.95         | 9.4    | 16.5   | 33.6   | 29.2   | 10.7   | 0.7    |
| U3 - cutoff = 0.75         | 14.4   | 20.1   | 33.8   | 25.1   | 5.5    | 1.1    |
| U3 - cutoff = 0.85         | 13.9   | 19.7   | 34.1   | 26     | 5.6    | 0.7    |
| U3 - cutoff = 0.95         | 13.8   | 21.4   | 32.1   | 27.5   | 4.6    | 0.7    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 3.7    | 4.4    | 7.8    | 7.9    | 6.1    | -      |
| U2 - cutoff = 0.85         | 3.9    | 4.4    | 8      | 7.5    | 6.1    | -      |
| U2 - cutoff = 0.95         | 3.9    | 4.7    | 7.9    | 7.7    | 5.7    | -      |
| U3 - cutoff = 0.75         | 3.7    | 4.4    | 7.8    | 7.9    | 6      | -      |
| U3 - cutoff = 0.85         | 3.9    | 4.4    | 8      | 7.5    | 6.1    | -      |
| U3 - cutoff = 0.95         | 4      | 4.6    | 8      | 7.7    | 5.6    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.3    | 2.9    | 2.5    | 3      | 2.1    | -      |
| U2 - cutoff = 0.85         | 0.4    | 2.9    | 2.5    | 2.8    | 2.1    | -      |
| U2 - cutoff = 0.95         | 0.3    | 3      | 2      | 2.7    | 1.1    | -      |
| U3 - cutoff = 0.75         | 0.3    | 2.9    | 2.5    | 2.9    | 2      | -      |
| U3 - cutoff = 0.85         | 0.4    | 2.9    | 2.5    | 2.8    | 2.1    | -      |
| U3 - cutoff = 0.95         | 0.3    | 2.9    | 2      | 2.7    | 1.1    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1054.7 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1055.6 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1150.2 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1055.5 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1056.5 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1149.7 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.



Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 33                |        |        |        |        |        |        |
| DLT rate                   | 0.01   | 0.02   | 0.09   | 0.32   | 0.38   |        |
| Efficacy rate              | 0.05   | 0.08   | 0.20   | 0.40   | 0.50   |        |
| Utility 2                  | 0      | 0.1    | 0.1    | 0.1    | 0.1    |        |
| Utility 3                  | 42.6   | 44     | 48.4   | 51.2   | 54.8   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 8.6    | 8      | 16.3   | 41.1   | 24.9   | 1.1    |
| U2 - cutoff = 0.85         | 7.9    | 8.8    | 15.7   | 42     | 25.5   | 0.1    |
| U2 - cutoff = 0.95         | 9.4    | 7.6    | 15.4   | 43.4   | 24.2   | 0      |
| U3 - cutoff = 0.75         | 13.6   | 9.8    | 13.1   | 40.6   | 21.9   | 1      |
| U3 - cutoff = 0.85         | 12     | 9.7    | 14.7   | 41.6   | 21.9   | 0.1    |
| U3 - cutoff = 0.95         | 14.5   | 9.6    | 14.6   | 42.6   | 18.8   | 0      |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 3.4    | 3.7    | 5.3    | 8.1    | 9.4    | -      |
| U2 - cutoff = 0.85         | 3.5    | 3.6    | 5.3    | 8.1    | 9.4    | -      |
| U2 - cutoff = 0.95         | 3.6    | 3.7    | 5.7    | 8.7    | 8.3    | -      |
| U3 - cutoff = 0.75         | 3.4    | 3.7    | 5.4    | 8.2    | 9.3    | -      |
| U3 - cutoff = 0.85         | 3.5    | 3.6    | 5.4    | 8.1    | 9.3    | -      |
| U3 - cutoff = 0.95         | 3.6    | 3.7    | 5.8    | 8.8    | 8.2    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.2    | 2.7    | 1.6    | 3.1    | 2.3    | -      |
| U2 - cutoff = 0.85         | 0.2    | 2.6    | 1.7    | 3.1    | 2.3    | -      |
| U2 - cutoff = 0.95         | 0.2    | 2.7    | 1.5    | 2.9    | 0.9    | -      |
| U3 - cutoff = 0.75         | 0.2    | 2.7    | 1.7    | 3.1    | 2.3    | -      |
| U3 - cutoff = 0.85         | 0.2    | 2.6    | 1.7    | 3      | 2.3    | -      |
| U3 - cutoff = 0.95         | 0.2    | 2.7    | 1.5    | 2.9    | 0.9    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1103.2 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1104.4 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1203.7 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1103.4 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1104.3 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1204.4 |
| Scenario 34                |        |        |        |        |        |        |
| DLT rate                   | 0.05   | 0.13   | 0.30   | 0.36   | 0.42   |        |
| Efficacy rate              | 0.05   | 0.10   | 0.30   | 0.32   | 0.35   |        |
| Utility 2                  | 0      | 0      | 0      | 0      | -0.1   |        |
| Utility 3                  | 41     | 40.8   | 46     | 44.8   | 44.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 13.1   | 12.8   | 44.2   | 13.3   | 2.7    | 13.9   |
| U2 - cutoff = 0.85         | 14.9   | 15     | 47.5   | 13.8   | 3      | 5.8    |
| U2 - cutoff = 0.95         | 17.3   | 17.8   | 47.2   | 13.2   | 2.4    | 2.1    |
| U3 - cutoff = 0.75         | 16.2   | 13.5   | 42.4   | 11.8   | 2      | 14.1   |
| U3 - cutoff = 0.85         | 19.8   | 17.3   | 43.5   | 11.5   | 2.1    | 5.8    |
| U3 - cutoff = 0.95         | 22.1   | 19     | 42.7   | 12.2   | 1.8    | 2.1    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.8    | 5.8    | 10.3   | 4.9    | 2.3    | -      |
| U2 - cutoff = 0.85         | 5.3    | 6.3    | 10.3   | 4.9    | 2.3    | -      |
| U2 - cutoff = 0.95         | 5.9    | 6.8    | 10     | 4.9    | 2      | -      |
| U3 - cutoff = 0.75         | 4.8    | 5.8    | 10.1   | 4.9    | 2.4    | -      |
| U3 - cutoff = 0.85         | 5.4    | 6.3    | 10.1   | 4.9    | 2.4    | -      |
| U3 - cutoff = 0.95         | 5.8    | 6.9    | 10     | 4.9    | 1.9    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.8    | 3.3    | 3.3    | 1.7    | 0.7    | -      |
| U2 - cutoff = 0.85         | 1      | 3.5    | 3.3    | 1.7    | 0.7    | -      |
| U2 - cutoff = 0.95         | 0.6    | 3.4    | 2.1    | 1.5    | 0.4    | -      |
| U3 - cutoff = 0.75         | 0.8    | 3.3    | 3.3    | 1.7    | 0.8    | -      |
| U3 - cutoff = 0.85         | 1.1    | 3.5    | 3.2    | 1.7    | 0.8    | -      |
| U3 - cutoff = 0.95         | 0.5    | 3.4    | 2      | 1.5    | 0.4    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1004.3 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1032.8 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1195.2 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1000.8 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1030   |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1200   |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 35                |        |        |        |        |        |        |
| DLT rate                   | 0.03   | 0.09   | 0.30   | 0.40   | 0.45   |        |
| Efficacy rate              | 0.20   | 0.45   | 0.45   | 0.45   | 0.45   |        |
| Utility 2                  | 0.2    | 0.4    | 0.1    | 0.1    | 0      |        |
| Utility 3                  | 50.8   | 63.4   | 55     | 51     | 49     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 11.2   | 44     | 32.9   | 10     | 1.5    | 0.4    |
| U2 - cutoff = 0.85         | 11.3   | 45.6   | 31.2   | 9.9    | 1.8    | 0.2    |
| U2 - cutoff = 0.95         | 12.2   | 44.9   | 31.1   | 10.1   | 1.7    | 0.1    |
| U3 - cutoff = 0.75         | 13.8   | 48.9   | 27.7   | 8.2    | 1.1    | 0.3    |
| U3 - cutoff = 0.85         | 14.2   | 49.3   | 27.2   | 8.1    | 1      | 0.2    |
| U3 - cutoff = 0.95         | 14.8   | 52     | 24.1   | 7.9    | 1.1    | 0.1    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.5    | 9      | 9.4    | 4.9    | 2.2    | -      |
| U2 - cutoff = 0.85         | 4.5    | 9.1    | 9.2    | 4.9    | 2.3    | -      |
| U2 - cutoff = 0.95         | 4.8    | 9.4    | 8.9    | 4.8    | 2.1    | -      |
| U3 - cutoff = 0.75         | 4.5    | 9      | 9.3    | 4.9    | 2.3    | -      |
| U3 - cutoff = 0.85         | 4.5    | 9      | 9.3    | 4.9    | 2.2    | -      |
| U3 - cutoff = 0.95         | 4.7    | 9.5    | 9      | 4.8    | 2.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.6    | 3.8    | 2.3    | 1.5    | 0.6    | -      |
| U2 - cutoff = 0.85         | 0.6    | 3.9    | 2.3    | 1.4    | 0.6    | -      |
| U2 - cutoff = 0.95         | 0.3    | 3.2    | 1.7    | 1.3    | 0.4    | -      |
| U3 - cutoff = 0.75         | 0.6    | 3.8    | 2.3    | 1.4    | 0.7    | -      |
| U3 - cutoff = 0.85         | 0.7    | 3.8    | 2.3    | 1.5    | 0.6    | -      |
| U3 - cutoff = 0.95         | 0.3    | 3.3    | 1.7    | 1.3    | 0.4    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1161.9 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1163.2 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1268.5 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1161.8 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1163.2 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1268   |
| Scenario 36                |        |        |        |        |        |        |
| DLT rate                   | 0.13   | 0.29   | 0.35   | 0.40   | 0.43   |        |
| Efficacy rate              | 0.05   | 0.20   | 0.22   | 0.30   | 0.35   |        |
| Utility 2                  | -0.1   | -0.1   | -0.1   | -0.1   | -0.1   |        |
| Utility 3                  | 37.8   | 40.4   | 39.2   | 42     | 43.8   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 11     | 28.3   | 9.9    | 2.6    | 0.9    | 47.3   |
| U2 - cutoff = 0.85         | 18.6   | 37.3   | 10.2   | 3      | 0.8    | 30.1   |
| U2 - cutoff = 0.95         | 23.7   | 45.1   | 10.3   | 2.8    | 0.9    | 17.3   |
| U3 - cutoff = 0.75         | 14.4   | 27.9   | 7.9    | 2.4    | 0.8    | 46.6   |
| U3 - cutoff = 0.85         | 21.9   | 36.4   | 8.7    | 2.2    | 0.8    | 30     |
| U3 - cutoff = 0.95         | 27.1   | 44.1   | 8.4    | 2.4    | 0.6    | 17.5   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 6.9    | 7.8    | 4.8    | 1.9    | 0.7    | -      |
| U2 - cutoff = 0.85         | 9      | 8.4    | 5      | 1.9    | 0.7    | -      |
| U2 - cutoff = 0.95         | 11.2   | 8.9    | 4.8    | 2      | 0.6    | -      |
| U3 - cutoff = 0.75         | 6.9    | 7.8    | 4.8    | 1.9    | 0.7    | -      |
| U3 - cutoff = 0.85         | 8.9    | 8.4    | 5      | 1.9    | 0.7    | -      |
| U3 - cutoff = 0.95         | 11.3   | 8.8    | 4.8    | 1.9    | 0.6    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.6    | 3.7    | 1.6    | 0.8    | 0.2    | -      |
| U2 - cutoff = 0.85         | 2.4    | 4      | 1.6    | 0.8    | 0.2    | -      |
| U2 - cutoff = 0.95         | 0.5    | 3.6    | 1      | 0.7    | 0.1    | -      |
| U3 - cutoff = 0.75         | 1.6    | 3.7    | 1.5    | 0.8    | 0.2    | -      |
| U3 - cutoff = 0.85         | 2.4    | 4      | 1.6    | 0.8    | 0.2    | -      |
| U3 - cutoff = 0.95         | 0.5    | 3.6    | 1      | 0.7    | 0.1    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 782.6  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 874.2  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1189.8 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 783    |
| U3 - cutoff = 0.85         |        |        |        |        |        | 875.5  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1187.2 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = two-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 37                |        |        |        |        |        |        |
| DLT rate                   | 0.08   | 0.30   | 0.40   | 0.45   | 0.50   |        |
| Efficacy rate              | 0.05   | 0.20   | 0.20   | 0.20   | 0.20   |        |
| Utility 2                  | 0      | -0.1   | -0.2   | -0.2   | -0.3   |        |
| Utility 3                  | 39.8   | 40     | 36     | 34     | 32     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 15.6   | 33.1   | 8.3    | 0.8    | 0.1    | 42.1   |
| U2 - cutoff = 0.85         | 23.4   | 45.4   | 10.4   | 0.9    | 0.1    | 19.8   |
| U2 - cutoff = 0.95         | 27.9   | 54.2   | 9.3    | 0.9    | 0.1    | 7.6    |
| U3 - cutoff = 0.75         | 18.2   | 32.7   | 6.6    | 0.6    | 0      | 41.9   |
| U3 - cutoff = 0.85         | 27.7   | 43.5   | 7.7    | 0.9    | 0      | 20.2   |
| U3 - cutoff = 0.95         | 31.4   | 52.2   | 7.7    | 0.6    | 0.1    | 8      |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 6.6    | 9.8    | 5.5    | 1.5    | 0.5    | -      |
| U2 - cutoff = 0.85         | 8.3    | 10.8   | 5.5    | 1.5    | 0.5    | -      |
| U2 - cutoff = 0.95         | 10     | 10.9   | 5.5    | 1.6    | 0.4    | -      |
| U3 - cutoff = 0.75         | 6.6    | 9.8    | 5.5    | 1.5    | 0.5    | -      |
| U3 - cutoff = 0.85         | 8.3    | 10.8   | 5.5    | 1.5    | 0.5    | -      |
| U3 - cutoff = 0.95         | 10.1   | 10.9   | 5.5    | 1.5    | 0.4    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.4    | 4.6    | 1.7    | 0.7    | 0.2    | -      |
| U2 - cutoff = 0.85         | 2.1    | 5      | 1.7    | 0.6    | 0.2    | -      |
| U2 - cutoff = 0.95         | 0.6    | 4.1    | 1.2    | 0.7    | 0.1    | -      |
| U3 - cutoff = 0.75         | 1.4    | 4.6    | 1.7    | 0.7    | 0.2    | -      |
| U3 - cutoff = 0.85         | 2.1    | 5      | 1.7    | 0.6    | 0.2    | -      |
| U3 - cutoff = 0.95         | 0.6    | 4.1    | 1.1    | 0.7    | 0.1    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 844.7  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 926.5  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1205.9 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 843.2  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 926.9  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1203.3 |

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 1                 |        |        |        |        |        |        |
| DLT rate                   | 0.30   | 0.35   | 0.40   | 0.45   | 0.50   |        |
| Efficacy rate              | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |        |
| Utility 2                  | -0.2   | -0.3   | -0.3   | -0.4   | -0.4   |        |
| Utility 3                  | 31     | 29.6   | 28.2   | 26.8   | 25.4   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 5.7    | 1.9    | 0.8    | 0      | 0      | 91.6   |
| U2 - cutoff = 0.85         | 14.6   | 4      | 0.9    | 0      | 0      | 80.5   |
| U2 - cutoff = 0.95         | 29.1   | 8.3    | 1.2    | 0      | 0      | 61.4   |
| U3 - cutoff = 0.75         | 5.9    | 1.9    | 0.5    | 0      | 0      | 91.7   |
| U3 - cutoff = 0.85         | 14.6   | 3.8    | 0.6    | 0      | 0      | 81     |
| U3 - cutoff = 0.95         | 29.3   | 8.1    | 0.6    | 0      | 0      | 62     |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 7.9    | 4.3    | 1.3    | 0.6    | 0.1    | -      |
| U2 - cutoff = 0.85         | 11.4   | 4.8    | 1.4    | 0.6    | 0.1    | -      |
| U2 - cutoff = 0.95         | 15.5   | 5.6    | 1.4    | 0.6    | 0.1    | -      |
| U3 - cutoff = 0.75         | 8      | 4.3    | 1.3    | 0.6    | 0.1    | -      |
| U3 - cutoff = 0.85         | 11.6   | 4.8    | 1.4    | 0.6    | 0.1    | -      |
| U3 - cutoff = 0.95         | 15.8   | 5.3    | 1.4    | 0.6    | 0.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.6    | 2.4    | 0.4    | 0.3    | 0      | -      |
| U2 - cutoff = 0.85         | 3.1    | 2.6    | 0.5    | 0.3    | 0      | -      |
| U2 - cutoff = 0.95         | 1      | 3.1    | 0.5    | 0.3    | 0      | -      |
| U3 - cutoff = 0.75         | 1.7    | 2.4    | 0.4    | 0.3    | 0      | -      |
| U3 - cutoff = 0.85         | 3.2    | 2.6    | 0.5    | 0.3    | 0      | -      |
| U3 - cutoff = 0.95         | 1.1    | 3      | 0.4    | 0.3    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 512.5  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 646.3  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1004   |
| U3 - cutoff = 0.75         |        |        |        |        |        | 515    |
| U3 - cutoff = 0.85         |        |        |        |        |        | 647.5  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1008.5 |
| Scenario 2                 |        |        |        |        |        |        |
| DLT rate                   | 0.15   | 0.20   | 0.30   | 0.35   | 0.45   |        |
| Efficacy rate              | 0.20   | 0.25   | 0.50   | 0.50   | 0.50   |        |
| Utility 2                  | 0.1    | 0.1    | 0.2    | 0.1    | 0.1    |        |
| Utility 3                  | 46     | 47     | 58     | 56     | 52     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 24.1   | 21.1   | 36.7   | 7.4    | 0.4    | 10.3   |
| U2 - cutoff = 0.85         | 27.5   | 24.9   | 37.1   | 7.4    | 0.4    | 2.7    |
| U2 - cutoff = 0.95         | 28.7   | 26.4   | 34.5   | 8.7    | 1      | 0.7    |
| U3 - cutoff = 0.75         | 29.1   | 22.3   | 32.4   | 5.8    | 0.4    | 10     |
| U3 - cutoff = 0.85         | 31.8   | 26.1   | 33.9   | 5.2    | 0.3    | 2.7    |
| U3 - cutoff = 0.95         | 33.3   | 27.6   | 31     | 7      | 0.5    | 0.7    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 9.8    | 7.8    | 8.1    | 2.3    | 0.2    | -      |
| U2 - cutoff = 0.85         | 10.5   | 8.3    | 8.3    | 2.3    | 0.2    | -      |
| U2 - cutoff = 0.95         | 10.4   | 8.9    | 7.9    | 2.5    | 0.3    | -      |
| U3 - cutoff = 0.75         | 10.3   | 8      | 7.7    | 2      | 0.2    | -      |
| U3 - cutoff = 0.85         | 11     | 8.6    | 7.9    | 1.9    | 0.2    | -      |
| U3 - cutoff = 0.95         | 11.1   | 8.8    | 7.5    | 2.3    | 0.2    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 2.4    | 3.5    | 2.3    | 0.7    | 0.1    | -      |
| U2 - cutoff = 0.85         | 2.7    | 3.7    | 2.4    | 0.7    | 0.1    | -      |
| U2 - cutoff = 0.95         | 0.7    | 3.7    | 1.7    | 0.7    | 0.1    | -      |
| U3 - cutoff = 0.75         | 2.6    | 3.6    | 2.2    | 0.6    | 0.1    | -      |
| U3 - cutoff = 0.85         | 2.9    | 3.8    | 2.3    | 0.6    | 0.1    | -      |
| U3 - cutoff = 0.95         | 0.7    | 3.7    | 1.6    | 0.6    | 0.1    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1058.2 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1102.7 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1267.2 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1053.4 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1096.6 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1278.7 |



Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 3                 |        |        |        |        |        |        |
| DLT rate                   | 0.15   | 0.20   | 0.30   | 0.35   | 0.45   |        |
| Efficacy rate              | 0.10   | 0.40   | 0.40   | 0.40   | 0.40   |        |
| Utility 2                  | -0.1   | 0.2    | 0.1    | 0.1    | -0.1   |        |
| Utility 3                  | 40     | 56     | 52     | 50     | 46     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 7.2    | 43.9   | 21.7   | 5.6    | 0.6    | 21     |
| U2 - cutoff = 0.85         | 9.5    | 50.2   | 23.3   | 6.1    | 0.5    | 10.4   |
| U2 - cutoff = 0.95         | 11.8   | 53.7   | 23.9   | 6.4    | 0.9    | 3.3    |
| U3 - cutoff = 0.75         | 7.9    | 46.6   | 20.4   | 3.5    | 0.4    | 21.2   |
| U3 - cutoff = 0.85         | 10     | 54.9   | 21     | 3.4    | 0.4    | 10.3   |
| U3 - cutoff = 0.95         | 13.2   | 57.5   | 20     | 5.2    | 0.7    | 3.4    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 6.4    | 11.4   | 6      | 2.3    | 0.3    | -      |
| U2 - cutoff = 0.85         | 7.5    | 12.4   | 6.1    | 2.3    | 0.3    | -      |
| U2 - cutoff = 0.95         | 8.2    | 12.6   | 6.2    | 2.3    | 0.4    | -      |
| U3 - cutoff = 0.75         | 6.6    | 12.1   | 5.5    | 2      | 0.3    | -      |
| U3 - cutoff = 0.85         | 7.7    | 13.1   | 5.6    | 2      | 0.3    | -      |
| U3 - cutoff = 0.95         | 8.3    | 13.3   | 5.7    | 2.1    | 0.3    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.1    | 4.7    | 1.8    | 0.8    | 0.1    | -      |
| U2 - cutoff = 0.85         | 1.5    | 5.1    | 1.8    | 0.8    | 0.1    | -      |
| U2 - cutoff = 0.95         | 0.4    | 4.3    | 1.7    | 0.8    | 0.2    | -      |
| U3 - cutoff = 0.75         | 1.1    | 5      | 1.6    | 0.7    | 0.1    | -      |
| U3 - cutoff = 0.85         | 1.6    | 5.3    | 1.6    | 0.7    | 0.1    | -      |
| U3 - cutoff = 0.95         | 0.5    | 4.2    | 1.5    | 0.7    | 0.1    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 983.8  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1057   |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1232.9 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 982    |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1053.7 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1247.8 |
| Scenario 4                 |        |        |        |        |        |        |
| DLT rate                   | 0.05   | 0.10   | 0.18   | 0.30   | 0.40   |        |
| Efficacy rate              | 0.05   | 0.10   | 0.25   | 0.50   | 0.55   |        |
| Utility 2                  | 0      | 0      | 0.1    | 0.2    | 0.1    |        |
| Utility 3                  | 41     | 42     | 47.8   | 58     | 57     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 6      | 7.1    | 21.9   | 51     | 8.3    | 5.7    |
| U2 - cutoff = 0.85         | 5.8    | 7.8    | 22.9   | 53.4   | 8.3    | 1.8    |
| U2 - cutoff = 0.95         | 6.6    | 9.2    | 21.9   | 52     | 10.1   | 0.2    |
| U3 - cutoff = 0.75         | 8.1    | 7.9    | 19.8   | 52.2   | 6      | 6      |
| U3 - cutoff = 0.85         | 9.7    | 7.9    | 21.2   | 53.6   | 5.7    | 1.9    |
| U3 - cutoff = 0.95         | 9.6    | 11.6   | 20.2   | 50.5   | 7.7    | 0.3    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 3.9    | 4.8    | 7.4    | 10.4   | 2.3    | -      |
| U2 - cutoff = 0.85         | 4.3    | 4.9    | 7.7    | 10.6   | 2.3    | -      |
| U2 - cutoff = 0.95         | 4.5    | 5.2    | 7.5    | 10.3   | 2.5    | -      |
| U3 - cutoff = 0.75         | 4.1    | 4.9    | 7.5    | 10.4   | 2      | -      |
| U3 - cutoff = 0.85         | 4.5    | 5.2    | 7.7    | 10.5   | 2      | -      |
| U3 - cutoff = 0.95         | 4.8    | 5.2    | 7.5    | 10.3   | 2.2    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.4    | 3.1    | 2.2    | 3.4    | 0.7    | -      |
| U2 - cutoff = 0.85         | 0.6    | 3.1    | 2.3    | 3.4    | 0.7    | -      |
| U2 - cutoff = 0.95         | 0.2    | 3.1    | 1.7    | 2.5    | 0.7    | -      |
| U3 - cutoff = 0.75         | 0.4    | 3.1    | 2.2    | 3.4    | 0.6    | -      |
| U3 - cutoff = 0.85         | 0.6    | 3.3    | 2.3    | 3.4    | 0.6    | -      |
| U3 - cutoff = 0.95         | 0.3    | 3.2    | 1.7    | 2.5    | 0.5    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1059.8 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1088.1 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1202.6 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1051.7 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1083.5 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1205.3 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 5                 |        |        |        |        |        |        |
| DLT rate                   | 0.07   | 0.12   | 0.17   | 0.22   | 0.30   |        |
| Efficacy rate              | 0.05   | 0.30   | 0.30   | 0.30   | 0.30   |        |
| Utility 2                  | 0      | 0.2    | 0.1    | 0.1    | 0      |        |
| Utility 3                  | 40.2   | 53.2   | 51.2   | 49.2   | 46     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 5      | 32.7   | 25.7   | 22.2   | 9      | 5.4    |
| U2 - cutoff = 0.85         | 4.9    | 34.6   | 26.1   | 23     | 9.5    | 1.9    |
| U2 - cutoff = 0.95         | 4.9    | 37.5   | 25.4   | 22.8   | 9      | 0.4    |
| U3 - cutoff = 0.75         | 6.2    | 39.6   | 24.7   | 18     | 6      | 5.5    |
| U3 - cutoff = 0.85         | 6.9    | 42.7   | 24.4   | 18.6   | 5.5    | 1.9    |
| U3 - cutoff = 0.95         | 7.3    | 43.6   | 25     | 17.6   | 6      | 0.5    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 3.8    | 8      | 7.2    | 6.8    | 3.3    | -      |
| U2 - cutoff = 0.85         | 4      | 8.1    | 7.4    | 6.9    | 3.2    | -      |
| U2 - cutoff = 0.95         | 4.2    | 8.3    | 7.3    | 6.9    | 3.3    | -      |
| U3 - cutoff = 0.75         | 3.9    | 8.6    | 7.3    | 6.4    | 2.7    | -      |
| U3 - cutoff = 0.85         | 4.2    | 8.8    | 7.5    | 6.5    | 2.7    | -      |
| U3 - cutoff = 0.95         | 4.4    | 9      | 7.4    | 6.4    | 2.7    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.2    | 4.1    | 2.3    | 2.6    | 1.2    | -      |
| U2 - cutoff = 0.85         | 0.4    | 4.1    | 2.4    | 2.6    | 1.2    | -      |
| U2 - cutoff = 0.95         | 0.2    | 3.4    | 2      | 2.2    | 1.1    | -      |
| U3 - cutoff = 0.75         | 0.3    | 4.3    | 2.4    | 2.5    | 1      | -      |
| U3 - cutoff = 0.85         | 0.4    | 4.4    | 2.5    | 2.5    | 1      | -      |
| U3 - cutoff = 0.95         | 0.2    | 3.4    | 2.1    | 2      | 0.8    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1021.2 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1043.7 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1162.9 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1019.7 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1041.9 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1176.3 |
| Scenario 6                 |        |        |        |        |        |        |
| DLT rate                   | 0.29   | 0.34   | 0.38   | 0.43   | 0.52   |        |
| Efficacy rate              | 0.30   | 0.32   | 0.35   | 0.38   | 0.40   |        |
| Utility 2                  | 0      | 0      | 0      | -0.1   | -0.1   |        |
| Utility 3                  | 46.4   | 45.6   | 45.8   | 45.6   | 43.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 59.9   | 12.3   | 2.1    | 0.5    | 0.1    | 25.1   |
| U2 - cutoff = 0.85         | 63.6   | 13.1   | 2      | 0.5    | 0.1    | 20.7   |
| U2 - cutoff = 0.95         | 66.4   | 15.3   | 1.7    | 0.8    | 0      | 15.9   |
| U3 - cutoff = 0.75         | 61.4   | 11.6   | 1.5    | 0.5    | 0.1    | 24.9   |
| U3 - cutoff = 0.85         | 66.3   | 11.6   | 1.2    | 0.5    | 0.1    | 20.3   |
| U3 - cutoff = 0.95         | 68.4   | 14.6   | 0.9    | 0.3    | 0.1    | 15.7   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 18.2   | 5.6    | 1.4    | 0.5    | 0.1    | -      |
| U2 - cutoff = 0.85         | 18.9   | 5.7    | 1.4    | 0.5    | 0.1    | -      |
| U2 - cutoff = 0.95         | 18.9   | 6.2    | 1.5    | 0.5    | 0.1    | -      |
| U3 - cutoff = 0.75         | 18.6   | 5.4    | 1.3    | 0.4    | 0.1    | -      |
| U3 - cutoff = 0.85         | 19.5   | 5.5    | 1.2    | 0.4    | 0.1    | -      |
| U3 - cutoff = 0.95         | 19.4   | 6.1    | 1.3    | 0.4    | 0.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.7    | 2.3    | 0.5    | 0.2    | 0      | -      |
| U2 - cutoff = 0.85         | 4.9    | 2.4    | 0.5    | 0.2    | 0      | -      |
| U2 - cutoff = 0.95         | 0.7    | 2.6    | 0.5    | 0.2    | 0      | -      |
| U3 - cutoff = 0.75         | 4.8    | 2.3    | 0.4    | 0.2    | 0      | -      |
| U3 - cutoff = 0.85         | 5.1    | 2.3    | 0.4    | 0.2    | 0      | -      |
| U3 - cutoff = 0.95         | 0.8    | 2.6    | 0.4    | 0.1    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1003   |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1031.9 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1283.4 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1001.7 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1033.9 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1287.4 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 7                 |        |        |        |        |        |        |
| DLT rate                   | 0.16   | 0.20   | 0.26   | 0.31   | 0.36   |        |
| Efficacy rate              | 0.20   | 0.50   | 0.50   | 0.50   | 0.50   |        |
| Utility 2                  | 0      | 0.3    | 0.2    | 0.2    | 0.1    |        |
| Utility 3                  | 45.6   | 62     | 59.6   | 57.6   | 55.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 12.7   | 52.8   | 19.8   | 3.8    | 0.6    | 10.3   |
| U2 - cutoff = 0.85         | 14.9   | 57.1   | 19.8   | 4.5    | 0.5    | 3.2    |
| U2 - cutoff = 0.95         | 12.9   | 57.7   | 21.9   | 5.9    | 0.3    | 1.2    |
| U3 - cutoff = 0.75         | 13.5   | 56.3   | 16.2   | 3.1    | 0.3    | 10.6   |
| U3 - cutoff = 0.85         | 16.6   | 60.9   | 15.4   | 3.4    | 0.3    | 3.4    |
| U3 - cutoff = 0.95         | 15.7   | 61.9   | 17.6   | 3.1    | 0.5    | 1.2    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 8.2    | 14.1   | 4.7    | 1.1    | 0.1    | -      |
| U2 - cutoff = 0.85         | 8.8    | 14.6   | 4.8    | 1.2    | 0.1    | -      |
| U2 - cutoff = 0.95         | 8.8    | 14.6   | 5      | 1.3    | 0.2    | -      |
| U3 - cutoff = 0.75         | 8.5    | 14.7   | 4      | 0.9    | 0.1    | -      |
| U3 - cutoff = 0.85         | 9.2    | 15.3   | 4      | 0.9    | 0.1    | -      |
| U3 - cutoff = 0.95         | 9.1    | 15.3   | 4.3    | 0.9    | 0.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.6    | 5.1    | 1.3    | 0.4    | 0      | -      |
| U2 - cutoff = 0.85         | 1.9    | 5.3    | 1.3    | 0.4    | 0      | -      |
| U2 - cutoff = 0.95         | 0.4    | 3.9    | 1.2    | 0.4    | 0.1    | -      |
| U3 - cutoff = 0.75         | 1.7    | 5.3    | 1.1    | 0.3    | 0      | -      |
| U3 - cutoff = 0.85         | 2      | 5.5    | 1.1    | 0.3    | 0      | -      |
| U3 - cutoff = 0.95         | 0.4    | 3.9    | 1      | 0.3    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1093   |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1137.2 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1324.7 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1088.6 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1134   |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1339.7 |
| Scenario 8                 |        |        |        |        |        |        |
| DLT rate                   | 0.32   | 0.37   | 0.42   | 0.48   | 0.52   |        |
| Efficacy rate              | 0.10   | 0.15   | 0.20   | 0.30   | 0.40   |        |
| Utility 2                  | -0.2   | -0.2   | -0.2   | -0.2   | -0.1   |        |
| Utility 3                  | 33.2   | 34.2   | 35.2   | 38.8   | 43.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 14.3   | 4.2    | 0.6    | 0.1    | 0      | 80.8   |
| U2 - cutoff = 0.85         | 29.6   | 5.5    | 0.4    | 0.2    | 0      | 64.3   |
| U2 - cutoff = 0.95         | 45.4   | 7      | 0.8    | 0.1    | 0.1    | 46.7   |
| U3 - cutoff = 0.75         | 15     | 3.8    | 0.5    | 0.1    | 0      | 80.6   |
| U3 - cutoff = 0.85         | 30.2   | 5      | 0.5    | 0.1    | 0      | 64.2   |
| U3 - cutoff = 0.95         | 45.8   | 6.7    | 0.6    | 0.1    | 0      | 46.9   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 10.3   | 4.6    | 1.1    | 0.4    | 0.1    | -      |
| U2 - cutoff = 0.85         | 14.1   | 4.9    | 1.1    | 0.4    | 0.1    | -      |
| U2 - cutoff = 0.95         | 17     | 5.3    | 1.2    | 0.4    | 0.1    | -      |
| U3 - cutoff = 0.75         | 10.4   | 4.5    | 1.1    | 0.4    | 0.1    | -      |
| U3 - cutoff = 0.85         | 14.3   | 4.8    | 1.1    | 0.4    | 0.1    | -      |
| U3 - cutoff = 0.95         | 17.3   | 5.1    | 1.1    | 0.4    | 0.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 2.4    | 2.3    | 0.4    | 0.2    | 0      | -      |
| U2 - cutoff = 0.85         | 3.9    | 2.5    | 0.4    | 0.2    | 0      | -      |
| U2 - cutoff = 0.95         | 0.9    | 2.7    | 0.4    | 0.2    | 0      | -      |
| U3 - cutoff = 0.75         | 2.5    | 2.3    | 0.4    | 0.2    | 0      | -      |
| U3 - cutoff = 0.85         | 4      | 2.4    | 0.4    | 0.2    | 0      | -      |
| U3 - cutoff = 0.95         | 0.9    | 2.6    | 0.4    | 0.2    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 608    |
| U2 - cutoff = 0.85         |        |        |        |        |        | 746.1  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1096.2 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 609.3  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 746.4  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1095.9 |

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 9                 |        |        |        |        |        |        |
| DLT rate                   | 0.15   | 0.18   | 0.22   | 0.26   | 0.29   |        |
| Efficacy rate              | 0.05   | 0.08   | 0.20   | 0.25   | 0.50   |        |
| Utility 2                  | -0.1   | -0.1   | 0      | 0      | 0.2    |        |
| Utility 3                  | 37     | 37.6   | 43.2   | 44.6   | 58.4   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 11.1   | 8.1    | 19.1   | 14.1   | 14.4   | 33.2   |
| U2 - cutoff = 0.85         | 15     | 13.9   | 22.4   | 15.3   | 14.7   | 18.7   |
| U2 - cutoff = 0.95         | 19.2   | 19.9   | 23.2   | 14.4   | 15.6   | 7.6    |
| U3 - cutoff = 0.75         | 12.9   | 9.9    | 17.2   | 13.8   | 12.4   | 33.8   |
| U3 - cutoff = 0.85         | 18.3   | 15.8   | 20.7   | 13.7   | 12.2   | 19.3   |
| U3 - cutoff = 0.95         | 23.6   | 24.1   | 18.9   | 13.3   | 12.2   | 7.8    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 5.7    | 5.3    | 5.5    | 4.8    | 3      | -      |
| U2 - cutoff = 0.85         | 7.5    | 6.2    | 6      | 4.9    | 3.1    | -      |
| U2 - cutoff = 0.95         | 8.5    | 7.1    | 5.8    | 4.7    | 3.2    | -      |
| U3 - cutoff = 0.75         | 5.9    | 5.4    | 5.4    | 4.7    | 2.8    | -      |
| U3 - cutoff = 0.85         | 7.7    | 6.3    | 5.9    | 4.9    | 2.8    | -      |
| U3 - cutoff = 0.95         | 9.2    | 6.8    | 5.7    | 4.6    | 3      | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1      | 3.2    | 1.8    | 1.9    | 1      | -      |
| U2 - cutoff = 0.85         | 1.7    | 3.5    | 2      | 1.9    | 1      | -      |
| U2 - cutoff = 0.95         | 0.6    | 4      | 1.7    | 1.6    | 0.8    | -      |
| U3 - cutoff = 0.75         | 1      | 3.2    | 1.8    | 1.9    | 0.9    | -      |
| U3 - cutoff = 0.85         | 1.8    | 3.6    | 2      | 1.9    | 0.9    | -      |
| U3 - cutoff = 0.95         | 0.7    | 3.9    | 1.6    | 1.6    | 0.7    | -      |
| Mean trial duration (days) |        |        |        |        |        |        |
| U2 - cutoff = 0.75         |        |        |        |        |        | Days   |
| U2 - cutoff = 0.85         |        |        |        |        |        | 851.3  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 954.1  |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1134.1 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 846.9  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 951.5  |
|                            |        |        |        |        |        | 1143.4 |
|                            |        |        |        |        |        |        |
| Scenario 10                |        |        |        |        |        |        |
| DLT rate                   | 0.23   | 0.28   | 0.32   | 0.39   | 0.43   |        |
| Efficacy rate              | 0.05   | 0.40   | 0.42   | 0.45   | 0.50   |        |
| Utility 2                  | -0.2   | 0.1    | 0.1    | 0.1    | 0.1    |        |
| Utility 3                  | 33.8   | 52.8   | 52.4   | 51.4   | 52.8   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.4    | 23.1   | 10.9   | 2.2    | 0.1    | 59.3   |
| U2 - cutoff = 0.85         | 10.6   | 28.5   | 12.3   | 2.3    | 0.1    | 46.2   |
| U2 - cutoff = 0.95         | 19.4   | 36.4   | 12.5   | 2.9    | 0.3    | 28.5   |
| U3 - cutoff = 0.75         | 4.5    | 24.4   | 9.9    | 1.7    | 0.1    | 59.4   |
| U3 - cutoff = 0.85         | 11     | 29.5   | 10.5   | 2      | 0.1    | 46.9   |
| U3 - cutoff = 0.95         | 20.8   | 36.3   | 10.9   | 2.4    | 0.3    | 29.3   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 7.2    | 8.3    | 3.3    | 1.1    | 0.2    | -      |
| U2 - cutoff = 0.85         | 9.6    | 9.2    | 3.4    | 1.1    | 0.2    | -      |
| U2 - cutoff = 0.95         | 12     | 10     | 3.7    | 1.2    | 0.2    | -      |
| U3 - cutoff = 0.75         | 7.3    | 8.5    | 3.2    | 1      | 0.1    | -      |
| U3 - cutoff = 0.85         | 9.7    | 9.3    | 3.3    | 1      | 0.1    | -      |
| U3 - cutoff = 0.95         | 12.1   | 10.2   | 3.5    | 1      | 0.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.2    | 3.7    | 0.9    | 0.4    | 0.1    | -      |
| U2 - cutoff = 0.85         | 2.2    | 4.1    | 1      | 0.4    | 0.1    | -      |
| U2 - cutoff = 0.95         | 0.6    | 4.2    | 0.9    | 0.4    | 0.1    | -      |
| U3 - cutoff = 0.75         | 1.2    | 3.8    | 0.9    | 0.4    | 0      | -      |
| U3 - cutoff = 0.85         | 2.3    | 4.1    | 1      | 0.4    | 0      | -      |
| U3 - cutoff = 0.95         | 0.6    | 4.1    | 0.8    | 0.3    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        |        |
| U2 - cutoff = 0.75         |        |        |        |        |        | Days   |
| U2 - cutoff = 0.85         |        |        |        |        |        | 753.3  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 865    |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1156.3 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 754.9  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 864    |
|                            |        |        |        |        |        | 1162.8 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.



Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 11                |        |        |        |        |        |        |
| DLT rate                   | 0.22   | 0.32   | 0.44   | 0.59   | 0.67   |        |
| Efficacy rate              | 0.20   | 0.20   | 0.20   | 0.20   | 0.20   |        |
| Utility 2                  | 0      | -0.1   | -0.2   | -0.4   | -0.5   |        |
| Utility 3                  | 43.2   | 39.2   | 34.4   | 28.4   | 25.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 54.1   | 15.3   | 1.1    | 0.1    | 0      | 29.4   |
| U2 - cutoff = 0.85         | 65.5   | 17.4   | 1.3    | 0      | 0      | 15.8   |
| U2 - cutoff = 0.95         | 68.1   | 21.6   | 2.1    | 0      | 0      | 8.2    |
| U3 - cutoff = 0.75         | 54.8   | 14.1   | 1.1    | 0.1    | 0      | 29.9   |
| U3 - cutoff = 0.85         | 66.6   | 16.7   | 1      | 0      | 0      | 15.7   |
| U3 - cutoff = 0.95         | 70.2   | 19.9   | 1.6    | 0      | 0      | 8.3    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 15.7   | 6.8    | 2.2    | 0.7    | 0      | -      |
| U2 - cutoff = 0.85         | 17.7   | 7.2    | 2.2    | 0.7    | 0      | -      |
| U2 - cutoff = 0.95         | 18.1   | 7.7    | 2.2    | 0.7    | 0      | -      |
| U3 - cutoff = 0.75         | 15.9   | 6.7    | 2      | 0.7    | 0      | -      |
| U3 - cutoff = 0.85         | 18     | 7.1    | 2.1    | 0.7    | 0      | -      |
| U3 - cutoff = 0.95         | 18.5   | 7.5    | 2.1    | 0.7    | 0      | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.4    | 3.2    | 0.7    | 0.3    | 0      | -      |
| U2 - cutoff = 0.85         | 5.2    | 3.4    | 0.7    | 0.3    | 0      | -      |
| U2 - cutoff = 0.95         | 1      | 3.7    | 0.7    | 0.3    | 0      | -      |
| U3 - cutoff = 0.75         | 4.5    | 3.2    | 0.6    | 0.3    | 0      | -      |
| U3 - cutoff = 0.85         | 5.3    | 3.4    | 0.6    | 0.3    | 0      | -      |
| U3 - cutoff = 0.95         | 1      | 3.6    | 0.7    | 0.3    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 925.4  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1005.3 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1276.6 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 920.8  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1003.5 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1281.3 |
| Scenario 12                |        |        |        |        |        |        |
| DLT rate                   | 0.06   | 0.09   | 0.18   | 0.29   | 0.44   |        |
| Efficacy rate              | 0.05   | 0.10   | 0.12   | 0.40   | 0.60   |        |
| Utility 2                  | 0      | 0      | -0.1   | 0.1    | 0.2    |        |
| Utility 3                  | 40.6   | 42.4   | 40     | 52.4   | 58.4   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 10.9   | 13.6   | 13.9   | 41.9   | 12.6   | 7.1    |
| U2 - cutoff = 0.85         | 11.5   | 14.1   | 13.4   | 46.4   | 12.2   | 2.4    |
| U2 - cutoff = 0.95         | 12.7   | 16.9   | 13.1   | 42.9   | 14.1   | 0.3    |
| U3 - cutoff = 0.75         | 15.1   | 13.9   | 11.1   | 40.7   | 11.1   | 8.1    |
| U3 - cutoff = 0.85         | 16.5   | 15.4   | 11.8   | 43     | 10.9   | 2.4    |
| U3 - cutoff = 0.95         | 16     | 17.8   | 12     | 40.6   | 13.2   | 0.5    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.3    | 5.5    | 5.9    | 9.2    | 3.6    | -      |
| U2 - cutoff = 0.85         | 4.7    | 5.8    | 6.3    | 9.4    | 3.6    | -      |
| U2 - cutoff = 0.95         | 5      | 6.1    | 6.2    | 8.8    | 3.9    | -      |
| U3 - cutoff = 0.75         | 4.5    | 5.6    | 5.8    | 9.1    | 3.4    | -      |
| U3 - cutoff = 0.85         | 5      | 6      | 6      | 9.3    | 3.4    | -      |
| U3 - cutoff = 0.95         | 5.6    | 6.1    | 6      | 8.8    | 3.6    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.5    | 3.4    | 1.8    | 3.2    | 1      | -      |
| U2 - cutoff = 0.85         | 0.7    | 3.6    | 1.9    | 3.3    | 1      | -      |
| U2 - cutoff = 0.95         | 0.4    | 3.4    | 1.7    | 2.4    | 1      | -      |
| U3 - cutoff = 0.75         | 0.6    | 3.5    | 1.7    | 3.2    | 1      | -      |
| U3 - cutoff = 0.85         | 0.9    | 3.6    | 1.8    | 3.3    | 1      | -      |
| U3 - cutoff = 0.95         | 0.5    | 3.4    | 1.7    | 2.3    | 0.9    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1017.7 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1060.5 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1165.7 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1013.4 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1057   |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1174.6 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 13                |        |        |        |        |        |        |
| DLT rate                   | 0.21   | 0.29   | 0.30   | 0.39   | 0.46   |        |
| Efficacy rate              | 0.05   | 0.30   | 0.32   | 0.35   | 0.40   |        |
| Utility 2                  | -0.2   | 0      | 0      | 0      | -0.1   |        |
| Utility 3                  | 34.6   | 46.4   | 47.2   | 45.4   | 45.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 6.3    | 22.8   | 9.1    | 3.5    | 0.5    | 57.8   |
| U2 - cutoff = 0.85         | 12     | 29     | 10     | 3.7    | 0.4    | 44.9   |
| U2 - cutoff = 0.95         | 21.9   | 37.9   | 11     | 3.8    | 0.4    | 25     |
| U3 - cutoff = 0.75         | 7.3    | 23.7   | 8.5    | 2.4    | 0.4    | 57.7   |
| U3 - cutoff = 0.85         | 13.2   | 29.8   | 9.9    | 2.4    | 0.4    | 44.3   |
| U3 - cutoff = 0.95         | 22.3   | 39.5   | 9.4    | 2.8    | 0.3    | 25.7   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 7.3    | 7.4    | 3.6    | 1.7    | 0.3    | -      |
| U2 - cutoff = 0.85         | 9.7    | 8.5    | 3.8    | 1.8    | 0.3    | -      |
| U2 - cutoff = 0.95         | 12     | 9.5    | 4      | 1.7    | 0.3    | -      |
| U3 - cutoff = 0.75         | 7.3    | 7.8    | 3.5    | 1.5    | 0.3    | -      |
| U3 - cutoff = 0.85         | 9.8    | 8.8    | 3.6    | 1.6    | 0.3    | -      |
| U3 - cutoff = 0.95         | 12.2   | 9.7    | 3.8    | 1.6    | 0.3    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.3    | 3.6    | 1.1    | 0.7    | 0.1    | -      |
| U2 - cutoff = 0.85         | 2.3    | 4      | 1.2    | 0.7    | 0.1    | -      |
| U2 - cutoff = 0.95         | 0.7    | 4.4    | 1      | 0.7    | 0.1    | -      |
| U3 - cutoff = 0.75         | 1.3    | 3.7    | 1.1    | 0.6    | 0.1    | -      |
| U3 - cutoff = 0.85         | 2.4    | 4.2    | 1.1    | 0.6    | 0.1    | -      |
| U3 - cutoff = 0.95         | 0.7    | 4.4    | 1      | 0.6    | 0.1    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 740    |
| U2 - cutoff = 0.85         |        |        |        |        |        | 859.7  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1138.1 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 741.6  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 856.3  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1143.8 |
|                            |        |        |        |        |        |        |
| Scenario 14                |        |        |        |        |        |        |
| DLT rate                   | 0.09   | 0.15   | 0.23   | 0.28   | 0.39   |        |
| Efficacy rate              | 0.05   | 0.08   | 0.12   | 0.30   | 0.40   |        |
| Utility 2                  | 0      | -0.1   | -0.1   | 0      | 0      |        |
| Utility 3                  | 39.4   | 38.8   | 38     | 46.8   | 48.4   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 14.3   | 12.9   | 17.2   | 26.8   | 8.2    | 20.6   |
| U2 - cutoff = 0.85         | 17.3   | 17.9   | 19.7   | 28.1   | 8.9    | 8.1    |
| U2 - cutoff = 0.95         | 20.9   | 22.4   | 19.6   | 26.4   | 9      | 1.8    |
| U3 - cutoff = 0.75         | 17.5   | 16.3   | 14.7   | 23.5   | 6.9    | 21.1   |
| U3 - cutoff = 0.85         | 22.7   | 21.7   | 15.2   | 25.7   | 6.5    | 8.2    |
| U3 - cutoff = 0.95         | 28.1   | 26     | 13.5   | 23.9   | 7      | 1.6    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 5.2    | 6      | 5.9    | 6.7    | 2.8    | -      |
| U2 - cutoff = 0.85         | 6.3    | 6.8    | 6.4    | 6.9    | 2.9    | -      |
| U2 - cutoff = 0.95         | 6.9    | 7.2    | 6.2    | 6.6    | 3      | -      |
| U3 - cutoff = 0.75         | 5.5    | 6.2    | 5.8    | 6.5    | 2.7    | -      |
| U3 - cutoff = 0.85         | 6.6    | 7      | 6.2    | 6.7    | 2.7    | -      |
| U3 - cutoff = 0.95         | 7.5    | 7      | 5.9    | 6.5    | 2.8    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.9    | 3.6    | 1.9    | 2.6    | 0.9    | -      |
| U2 - cutoff = 0.85         | 1.4    | 3.9    | 2.1    | 2.7    | 1      | -      |
| U2 - cutoff = 0.95         | 0.7    | 3.9    | 1.8    | 2      | 0.8    | -      |
| U3 - cutoff = 0.75         | 1      | 3.7    | 1.8    | 2.5    | 0.9    | -      |
| U3 - cutoff = 0.85         | 1.5    | 4      | 2      | 2.6    | 0.9    | -      |
| U3 - cutoff = 0.95         | 0.8    | 3.8    | 1.7    | 2      | 0.7    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 922.6  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1001.5 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1139.6 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 918.5  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 997.9  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1145.6 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 15                |        |        |        |        |        |        |
| DLT rate                   | 0.13   | 0.20   | 0.32   | 0.39   | 0.46   |        |
| Efficacy rate              | 0.10   | 0.10   | 0.10   | 0.10   | 0.10   |        |
| Utility 2                  | 0      | -0.1   | -0.2   | -0.3   | -0.4   |        |
| Utility 3                  | 40.8   | 38     | 33.2   | 30.4   | 27.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 31.8   | 20.4   | 11.9   | 2.7    | 0.4    | 32.8   |
| U2 - cutoff = 0.85         | 40.6   | 29.8   | 11.4   | 4.2    | 0.3    | 13.7   |
| U2 - cutoff = 0.95         | 48.5   | 35     | 11     | 2.7    | 0.1    | 2.8    |
| U3 - cutoff = 0.75         | 33.2   | 19.8   | 10.5   | 3      | 0.4    | 33.1   |
| U3 - cutoff = 0.85         | 44.8   | 30.7   | 7.4    | 2.8    | 0.3    | 14     |
| U3 - cutoff = 0.95         | 53.5   | 33.6   | 7.8    | 1.9    | 0.1    | 3.1    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 8.7    | 7.9    | 5      | 3      | 0.8    | -      |
| U2 - cutoff = 0.85         | 10.7   | 8.7    | 5.3    | 3      | 0.8    | -      |
| U2 - cutoff = 0.95         | 11.4   | 9.3    | 5.3    | 3      | 0.8    | -      |
| U3 - cutoff = 0.75         | 9      | 7.7    | 4.9    | 2.9    | 0.8    | -      |
| U3 - cutoff = 0.85         | 11.1   | 8.6    | 5.1    | 2.9    | 0.8    | -      |
| U3 - cutoff = 0.95         | 12.4   | 8.8    | 4.9    | 2.8    | 0.8    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 2.2    | 4.1    | 1.7    | 1.3    | 0.3    | -      |
| U2 - cutoff = 0.85         | 3.1    | 4.4    | 1.8    | 1.3    | 0.3    | -      |
| U2 - cutoff = 0.95         | 1.1    | 4.4    | 1.8    | 1.2    | 0.3    | -      |
| U3 - cutoff = 0.75         | 2.4    | 4      | 1.6    | 1.2    | 0.3    | -      |
| U3 - cutoff = 0.85         | 3.3    | 4.4    | 1.7    | 1.2    | 0.3    | -      |
| U3 - cutoff = 0.95         | 1.3    | 4.2    | 1.6    | 1.2    | 0.3    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 859.6  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 960.3  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1150.2 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 857.2  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 959.7  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1163.2 |
| Scenario 16                |        |        |        |        |        |        |
| DLT rate                   | 0.02   | 0.06   | 0.09   | 0.14   | 0.26   |        |
| Efficacy rate              | 0.20   | 0.20   | 0.20   | 0.20   | 0.20   |        |
| Utility 2                  | 0.2    | 0.1    | 0.1    | 0.1    | -0.1   |        |
| Utility 3                  | 51.2   | 49.6   | 48.4   | 46.4   | 41.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 34.6   | 29.7   | 15     | 13.4   | 7.1    | 0.2    |
| U2 - cutoff = 0.85         | 35.1   | 30.2   | 15.5   | 12.8   | 6.4    | 0      |
| U2 - cutoff = 0.95         | 35.4   | 28.5   | 15.4   | 13.4   | 7.3    | 0      |
| U3 - cutoff = 0.75         | 40.4   | 31.3   | 13.3   | 10.4   | 4.4    | 0.2    |
| U3 - cutoff = 0.85         | 39.6   | 32     | 13.8   | 10.4   | 4.2    | 0      |
| U3 - cutoff = 0.95         | 41.2   | 31.3   | 13.9   | 8.9    | 4.6    | 0      |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.4    | 5.3    | 6      | 8.2    | 6      | -      |
| U2 - cutoff = 0.85         | 4.4    | 5.3    | 6.1    | 8.2    | 6      | -      |
| U2 - cutoff = 0.95         | 4.4    | 5.5    | 6.1    | 8.2    | 5.8    | -      |
| U3 - cutoff = 0.75         | 4.8    | 5.8    | 6.2    | 7.9    | 5.3    | -      |
| U3 - cutoff = 0.85         | 4.7    | 5.8    | 6.2    | 7.9    | 5.4    | -      |
| U3 - cutoff = 0.95         | 4.5    | 5.9    | 6.3    | 8.1    | 5.2    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.6    | 3      | 2.3    | 3.5    | 2.3    | -      |
| U2 - cutoff = 0.85         | 0.6    | 3      | 2.3    | 3.5    | 2.3    | -      |
| U2 - cutoff = 0.95         | 0.2    | 2.8    | 2.2    | 2.9    | 1.5    | -      |
| U3 - cutoff = 0.75         | 0.7    | 3.2    | 2.3    | 3.4    | 2      | -      |
| U3 - cutoff = 0.85         | 0.7    | 3.2    | 2.3    | 3.4    | 2.1    | -      |
| U3 - cutoff = 0.95         | 0.2    | 2.9    | 2.2    | 2.9    | 1.4    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1005.7 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1007.8 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1120.7 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1003.2 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1005.5 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1120.8 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 17                |        |        |        |        |        |        |
| DLT rate                   | 0.11   | 0.23   | 0.39   | 0.58   | 0.76   |        |
| Efficacy rate              | 0.30   | 0.50   | 0.50   | 0.50   | 0.50   |        |
| Utility 2                  | 0.2    | 0.3    | 0.1    | -0.1   | -0.3   |        |
| Utility 3                  | 53.6   | 60.8   | 54.4   | 46.8   | 39.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 19.8   | 66     | 10.4   | 0.7    | 0      | 3.1    |
| U2 - cutoff = 0.85         | 20.4   | 68.5   | 10.2   | 0.5    | 0      | 0.4    |
| U2 - cutoff = 0.95         | 17.7   | 68.4   | 13.1   | 0.7    | 0      | 0.1    |
| U3 - cutoff = 0.75         | 22.8   | 66.5   | 7      | 0.3    | 0      | 3.4    |
| U3 - cutoff = 0.85         | 23.6   | 68.7   | 7      | 0.3    | 0      | 0.4    |
| U3 - cutoff = 0.95         | 19.3   | 69.6   | 10.4   | 0.5    | 0      | 0.2    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 9.9    | 15.8   | 3.5    | 0.4    | 0      | -      |
| U2 - cutoff = 0.85         | 10.1   | 16.2   | 3.3    | 0.4    | 0      | -      |
| U2 - cutoff = 0.95         | 9.6    | 16     | 4      | 0.4    | 0      | -      |
| U3 - cutoff = 0.75         | 10.5   | 16.1   | 2.5    | 0.3    | 0      | -      |
| U3 - cutoff = 0.85         | 10.8   | 16.4   | 2.5    | 0.3    | 0      | -      |
| U3 - cutoff = 0.95         | 10.1   | 16.4   | 3.2    | 0.2    | 0      | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 2.2    | 5.2    | 1      | 0.1    | 0      | -      |
| U2 - cutoff = 0.85         | 2.3    | 5.4    | 1      | 0.1    | 0      | -      |
| U2 - cutoff = 0.95         | 0.5    | 4.1    | 1.1    | 0.2    | 0      | -      |
| U3 - cutoff = 0.75         | 2.4    | 5.4    | 0.7    | 0.1    | 0      | -      |
| U3 - cutoff = 0.85         | 2.5    | 5.5    | 0.7    | 0.1    | 0      | -      |
| U3 - cutoff = 0.95         | 0.6    | 4.1    | 0.8    | 0.1    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        |        |
| U2 - cutoff = 0.75         |        |        |        |        |        | Days   |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1155.3 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1171   |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1331.1 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1147.6 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1165.5 |
|                            |        |        |        |        |        | 1348.6 |
|                            |        |        |        |        |        |        |
| Scenario 18                |        |        |        |        |        |        |
| DLT rate                   | 0.02   | 0.04   | 0.10   | 0.20   | 0.35   |        |
| Efficacy rate              | 0.05   | 0.10   | 0.15   | 0.20   | 0.50   |        |
| Utility 2                  | 0      | 0.1    | 0.1    | 0      | 0.1    |        |
| Utility 3                  | 42.2   | 44.4   | 45     | 44     | 56     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 8.9    | 10.8   | 11.4   | 18.9   | 49.4   | 0.6    |
| U2 - cutoff = 0.85         | 9.2    | 10.8   | 11     | 19     | 49.9   | 0.1    |
| U2 - cutoff = 0.95         | 9.6    | 11.6   | 9.9    | 20.4   | 48.5   | 0      |
| U3 - cutoff = 0.75         | 14.2   | 12.2   | 10.6   | 17.5   | 44.9   | 0.6    |
| U3 - cutoff = 0.85         | 14.7   | 13.3   | 10.4   | 17.6   | 43.9   | 0.1    |
| U3 - cutoff = 0.95         | 17     | 16.9   | 9.3    | 15.8   | 41.1   | 0      |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 3.3    | 4.1    | 5.3    | 7.4    | 9.7    | -      |
| U2 - cutoff = 0.85         | 3.4    | 4.1    | 5.3    | 7.4    | 9.8    | -      |
| U2 - cutoff = 0.95         | 3.4    | 4.3    | 5.4    | 7.4    | 9.5    | -      |
| U3 - cutoff = 0.75         | 3.4    | 4.2    | 5.4    | 7.6    | 9.2    | -      |
| U3 - cutoff = 0.85         | 3.5    | 4.2    | 5.4    | 7.6    | 9.2    | -      |
| U3 - cutoff = 0.95         | 3.5    | 4.4    | 5.4    | 7.5    | 9.2    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.1    | 2.8    | 1.8    | 3      | 2.9    | -      |
| U2 - cutoff = 0.85         | 0.2    | 2.8    | 1.8    | 3      | 2.9    | -      |
| U2 - cutoff = 0.95         | 0.1    | 2.7    | 1.5    | 2.6    | 1.7    | -      |
| U3 - cutoff = 0.75         | 0.2    | 2.9    | 1.9    | 3.1    | 2.7    | -      |
| U3 - cutoff = 0.85         | 0.2    | 2.9    | 1.9    | 3.1    | 2.7    | -      |
| U3 - cutoff = 0.95         | 0.2    | 2.8    | 1.6    | 2.5    | 1.7    | -      |
| Mean trial duration (days) |        |        |        |        |        |        |
| U2 - cutoff = 0.75         |        |        |        |        |        | Days   |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1050.9 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1057.1 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1175.4 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1046.7 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1052.2 |
|                            |        |        |        |        |        | 1172.8 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.



Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 19                |        |        |        |        |        |        |
| DLT rate                   | 0.11   | 0.26   | 0.39   | 0.60   | 0.77   |        |
| Efficacy rate              | 0.08   | 0.30   | 0.32   | 0.35   | 0.40   |        |
| Utility 2                  | 0      | 0      | -0.1   | -0.2   | -0.4   |        |
| Utility 3                  | 40.4   | 47.6   | 43.6   | 37     | 33.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 11.1   | 44.5   | 12.1   | 0.4    | 0      | 31.9   |
| U2 - cutoff = 0.85         | 17     | 55     | 12.5   | 0.3    | 0      | 15.2   |
| U2 - cutoff = 0.95         | 20.7   | 59.7   | 14.8   | 0.6    | 0      | 4.2    |
| U3 - cutoff = 0.75         | 13.4   | 42.9   | 10.9   | 0.4    | 0      | 32.4   |
| U3 - cutoff = 0.85         | 18.2   | 54.5   | 10.9   | 0.3    | 0      | 16.1   |
| U3 - cutoff = 0.95         | 24.2   | 59.4   | 11.7   | 0.3    | 0      | 4.4    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 7.2    | 11.5   | 5.1    | 1.4    | 0.1    | -      |
| U2 - cutoff = 0.85         | 8.8    | 12.9   | 5.3    | 1.4    | 0.1    | -      |
| U2 - cutoff = 0.95         | 9.5    | 13.3   | 5.3    | 1.4    | 0.1    | -      |
| U3 - cutoff = 0.75         | 7.4    | 11.7   | 4.8    | 1.3    | 0.1    | -      |
| U3 - cutoff = 0.85         | 9      | 13     | 5.1    | 1.3    | 0.1    | -      |
| U3 - cutoff = 0.95         | 9.8    | 13.5   | 5      | 1.3    | 0.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.4    | 5.1    | 1.5    | 0.6    | 0      | -      |
| U2 - cutoff = 0.85         | 2      | 5.6    | 1.6    | 0.6    | 0      | -      |
| U2 - cutoff = 0.95         | 0.8    | 5      | 1.6    | 0.6    | 0      | -      |
| U3 - cutoff = 0.75         | 1.5    | 5.2    | 1.4    | 0.5    | 0      | -      |
| U3 - cutoff = 0.85         | 2.1    | 5.7    | 1.5    | 0.5    | 0      | -      |
| U3 - cutoff = 0.95         | 0.8    | 4.9    | 1.5    | 0.6    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 916    |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1017.3 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1192.4 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 914.1  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1012.5 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1203.2 |
|                            |        |        |        |        |        |        |
| Scenario 20                |        |        |        |        |        |        |
| DLT rate                   | 0.07   | 0.14   | 0.31   | 0.41   | 0.52   |        |
| Efficacy rate              | 0.05   | 0.25   | 0.25   | 0.25   | 0.25   |        |
| Utility 2                  | 0      | 0.1    | -0.1   | -0.2   | -0.3   |        |
| Utility 3                  | 40.2   | 49.4   | 42.6   | 38.6   | 34.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 8.3    | 47     | 26.8   | 7.5    | 0.6    | 9.8    |
| U2 - cutoff = 0.85         | 9.2    | 51.4   | 28.2   | 8.1    | 0.6    | 2.5    |
| U2 - cutoff = 0.95         | 10.3   | 50.9   | 30.2   | 7.4    | 0.6    | 0.6    |
| U3 - cutoff = 0.75         | 11.2   | 49.5   | 22.5   | 6      | 0.5    | 10.3   |
| U3 - cutoff = 0.85         | 11     | 56.4   | 23.1   | 6.4    | 0.5    | 2.6    |
| U3 - cutoff = 0.95         | 13.6   | 54     | 24.3   | 7      | 0.6    | 0.6    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.7    | 10.5   | 8      | 4.1    | 1      | -      |
| U2 - cutoff = 0.85         | 5.1    | 11.2   | 8.2    | 4.2    | 1      | -      |
| U2 - cutoff = 0.95         | 5.3    | 11.2   | 8.3    | 4.1    | 1      | -      |
| U3 - cutoff = 0.75         | 5      | 11     | 7.7    | 3.8    | 0.9    | -      |
| U3 - cutoff = 0.85         | 5.4    | 11.7   | 7.8    | 3.8    | 0.9    | -      |
| U3 - cutoff = 0.95         | 5.7    | 11.6   | 7.8    | 3.9    | 0.9    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.6    | 5      | 2.6    | 1.6    | 0.4    | -      |
| U2 - cutoff = 0.85         | 0.8    | 5.3    | 2.6    | 1.6    | 0.4    | -      |
| U2 - cutoff = 0.95         | 0.4    | 4.1    | 2.4    | 1.5    | 0.4    | -      |
| U3 - cutoff = 0.75         | 0.8    | 5.2    | 2.4    | 1.5    | 0.3    | -      |
| U3 - cutoff = 0.85         | 0.9    | 5.5    | 2.5    | 1.5    | 0.3    | -      |
| U3 - cutoff = 0.95         | 0.5    | 4.1    | 2.3    | 1.5    | 0.3    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1002   |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1043.4 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1161.8 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 998.7  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1039.6 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1167.1 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 21                |        |        |        |        |        |        |
| DLT rate                   | 0.37   | 0.56   | 0.69   | 0.83   | 0.88   |        |
| Efficacy rate              | 0.50   | 0.50   | 0.50   | 0.50   | 0.50   |        |
| Utility 2                  | 0.1    | -0.1   | -0.2   | -0.3   | -0.4   |        |
| Utility 3                  | 55.2   | 47.6   | 42.4   | 36.8   | 34.8   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 44.8   | 0.3    | 0      | 0      | 0      | 54.9   |
| U2 - cutoff = 0.85         | 45.4   | 0.1    | 0      | 0      | 0      | 54.5   |
| U2 - cutoff = 0.95         | 53.8   | 1.2    | 0      | 0      | 0      | 45     |
| U3 - cutoff = 0.75         | 44.9   | 0.1    | 0      | 0      | 0      | 55     |
| U3 - cutoff = 0.85         | 45.5   | 0.1    | 0      | 0      | 0      | 54.4   |
| U3 - cutoff = 0.95         | 52.7   | 1      | 0      | 0      | 0      | 46.3   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 19.1   | 1.9    | 0      | 0      | 0      | -      |
| U2 - cutoff = 0.85         | 19.2   | 1.9    | 0      | 0      | 0      | -      |
| U2 - cutoff = 0.95         | 19.8   | 2.3    | 0      | 0      | 0      | -      |
| U3 - cutoff = 0.75         | 19.1   | 1.8    | 0      | 0      | 0      | -      |
| U3 - cutoff = 0.85         | 19.3   | 1.8    | 0      | 0      | 0      | -      |
| U3 - cutoff = 0.95         | 19.8   | 2.3    | 0      | 0      | 0      | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 3.7    | 0.8    | 0      | 0      | 0      | -      |
| U2 - cutoff = 0.85         | 3.7    | 0.8    | 0      | 0      | 0      | -      |
| U2 - cutoff = 0.95         | 0.2    | 1.2    | 0      | 0      | 0      | -      |
| U3 - cutoff = 0.75         | 3.7    | 0.8    | 0      | 0      | 0      | -      |
| U3 - cutoff = 0.85         | 3.7    | 0.8    | 0      | 0      | 0      | -      |
| U3 - cutoff = 0.95         | 0.2    | 1.1    | 0      | 0      | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 912.2  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 914.6  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1147.9 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 908.4  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 914.7  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1144.3 |
| Scenario 22                |        |        |        |        |        |        |
| DLT rate                   | 0.07   | 0.27   | 0.50   | 0.56   | 0.78   |        |
| Efficacy rate              | 0.08   | 0.25   | 0.25   | 0.50   | 0.50   |        |
| Utility 2                  | 0      | 0      | -0.2   | -0.1   | -0.3   |        |
| Utility 3                  | 42     | 44.2   | 35     | 47.6   | 38.8   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 13.1   | 48.5   | 6.5    | 1      | 0      | 30.9   |
| U2 - cutoff = 0.85         | 17.5   | 60.4   | 8      | 0.9    | 0      | 13.2   |
| U2 - cutoff = 0.95         | 24.1   | 63.1   | 9.3    | 1.3    | 0      | 2.1    |
| U3 - cutoff = 0.75         | 13.5   | 48.1   | 6.1    | 1      | 0      | 31.3   |
| U3 - cutoff = 0.85         | 18.4   | 61.3   | 6.4    | 1.2    | 0      | 12.7   |
| U3 - cutoff = 0.95         | 25.7   | 61.3   | 8.8    | 1.6    | 0      | 2.6    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 7.5    | 12.2   | 4.7    | 1.4    | 0.1    | -      |
| U2 - cutoff = 0.85         | 8.9    | 13.6   | 4.9    | 1.5    | 0.1    | -      |
| U2 - cutoff = 0.95         | 9.6    | 13.8   | 4.9    | 1.5    | 0.1    | -      |
| U3 - cutoff = 0.75         | 7.8    | 12.3   | 4.4    | 1.3    | 0      | -      |
| U3 - cutoff = 0.85         | 9.2    | 13.7   | 4.6    | 1.4    | 0      | -      |
| U3 - cutoff = 0.95         | 10.1   | 13.7   | 4.6    | 1.4    | 0.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.5    | 5.5    | 1.4    | 0.6    | 0      | -      |
| U2 - cutoff = 0.85         | 2.1    | 6      | 1.5    | 0.6    | 0      | -      |
| U2 - cutoff = 0.95         | 1      | 5.1    | 1.6    | 0.6    | 0      | -      |
| U3 - cutoff = 0.75         | 1.6    | 5.6    | 1.3    | 0.5    | 0      | -      |
| U3 - cutoff = 0.85         | 2.3    | 6.1    | 1.4    | 0.5    | 0      | -      |
| U3 - cutoff = 0.95         | 1.1    | 5      | 1.6    | 0.6    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 927.4  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1025.1 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1187.5 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 921.5  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1021.1 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1190.3 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 23                |        |        |        |        |        |        |
| DLT rate                   | 0.32   | 0.55   | 0.80   | 0.94   | 0.97   |        |
| Efficacy rate              | 0.05   | 0.05   | 0.05   | 0.05   | 0.05   |        |
| Utility 2                  | -0.3   | -0.5   | -0.8   | -0.9   | -0.9   |        |
| Utility 3                  | 30.2   | 21     | 11     | 5.4    | 4.2    |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.9    | 0.1    | 0      | 0      | 0      | 98     |
| U2 - cutoff = 0.85         | 7      | 0.4    | 0      | 0      | 0      | 92.6   |
| U2 - cutoff = 0.95         | 17.1   | 2.1    | 0      | 0      | 0      | 80.8   |
| U3 - cutoff = 0.75         | 1.9    | 0.1    | 0      | 0      | 0      | 98     |
| U3 - cutoff = 0.85         | 6.4    | 0.6    | 0      | 0      | 0      | 93     |
| U3 - cutoff = 0.95         | 17.3   | 1.7    | 0      | 0      | 0      | 81     |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 7.7    | 3.4    | 0.3    | 0      | 0      | -      |
| U2 - cutoff = 0.85         | 11     | 3.5    | 0.3    | 0      | 0      | -      |
| U2 - cutoff = 0.95         | 15.3   | 3.9    | 0.3    | 0      | 0      | -      |
| U3 - cutoff = 0.75         | 7.7    | 3.4    | 0.3    | 0      | 0      | -      |
| U3 - cutoff = 0.85         | 10.9   | 3.5    | 0.3    | 0      | 0      | -      |
| U3 - cutoff = 0.95         | 15.3   | 3.8    | 0.3    | 0      | 0      | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.4    | 1.9    | 0.1    | 0      | 0      | -      |
| U2 - cutoff = 0.85         | 2.9    | 2      | 0.1    | 0      | 0      | -      |
| U2 - cutoff = 0.95         | 0.6    | 2.3    | 0.1    | 0      | 0      | -      |
| U3 - cutoff = 0.75         | 1.4    | 1.9    | 0.1    | 0      | 0      | -      |
| U3 - cutoff = 0.85         | 2.9    | 2      | 0.1    | 0      | 0      | -      |
| U3 - cutoff = 0.95         | 0.6    | 2.3    | 0.1    | 0      | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 425.9  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 531.3  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 908.4  |
| U3 - cutoff = 0.75         |        |        |        |        |        | 426    |
| U3 - cutoff = 0.85         |        |        |        |        |        | 529.8  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 905.8  |
| Scenario 24                |        |        |        |        |        |        |
| DLT rate                   | 0.01   | 0.02   | 0.05   | 0.10   | 0.25   |        |
| Efficacy rate              | 0.05   | 0.08   | 0.20   | 0.50   | 0.50   |        |
| Utility 2                  | 0      | 0.1    | 0.1    | 0.4    | 0.2    |        |
| Utility 3                  | 42.6   | 44     | 50     | 66     | 60     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.6    | 0.8    | 1.8    | 61.5   | 35.3   | 0      |
| U2 - cutoff = 0.85         | 0.6    | 0.9    | 2      | 60.6   | 35.9   | 0      |
| U2 - cutoff = 0.95         | 1.2    | 0.8    | 2.8    | 54.4   | 40.9   | 0      |
| U3 - cutoff = 0.75         | 1.4    | 1.2    | 3      | 64.6   | 29.8   | 0      |
| U3 - cutoff = 0.85         | 1.5    | 1.4    | 3.2    | 63.9   | 30     | 0      |
| U3 - cutoff = 0.95         | 2.1    | 2.6    | 2.9    | 56.6   | 35.7   | 0      |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 3.1    | 3.2    | 3.8    | 13.7   | 6.3    | -      |
| U2 - cutoff = 0.85         | 3.1    | 3.2    | 3.8    | 13.5   | 6.4    | -      |
| U2 - cutoff = 0.95         | 3.1    | 3.2    | 3.9    | 12.9   | 6.9    | -      |
| U3 - cutoff = 0.75         | 3.1    | 3.2    | 3.9    | 14.1   | 5.7    | -      |
| U3 - cutoff = 0.85         | 3.1    | 3.2    | 3.9    | 14.1   | 5.7    | -      |
| U3 - cutoff = 0.95         | 3.1    | 3.2    | 3.9    | 13.4   | 6.4    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0      | 2.4    | 1.2    | 4.7    | 1.9    | -      |
| U2 - cutoff = 0.85         | 0      | 2.4    | 1.2    | 4.7    | 1.9    | -      |
| U2 - cutoff = 0.95         | 0      | 2.4    | 1      | 2.4    | 1.5    | -      |
| U3 - cutoff = 0.75         | 0      | 2.5    | 1.2    | 4.9    | 1.7    | -      |
| U3 - cutoff = 0.85         | 0      | 2.5    | 1.2    | 4.9    | 1.7    | -      |
| U3 - cutoff = 0.95         | 0      | 2.4    | 1      | 2.4    | 1.4    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1089.5 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1088.5 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1246.6 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1086.5 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1086.9 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1252.5 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 25                |        |        |        |        |        |        |
| DLT rate                   | 0.09   | 0.18   | 0.37   | 0.63   | 0.77   |        |
| Efficacy rate              | 0.20   | 0.25   | 0.40   | 0.45   | 0.50   |        |
| Utility 2                  | 0.1    | 0.1    | 0      | -0.2   | -0.3   |        |
| Utility 3                  | 48.4   | 47.8   | 49.2   | 41.8   | 39.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 26.9   | 34.9   | 30.6   | 1.7    | 0      | 5.9    |
| U2 - cutoff = 0.85         | 27.5   | 37.7   | 31.9   | 2.1    | 0      | 0.8    |
| U2 - cutoff = 0.95         | 26.6   | 38.8   | 31.1   | 3.2    | 0      | 0.3    |
| U3 - cutoff = 0.75         | 30.8   | 35.1   | 26.9   | 1.3    | 0      | 5.9    |
| U3 - cutoff = 0.85         | 32.4   | 38     | 27     | 1.7    | 0      | 0.9    |
| U3 - cutoff = 0.95         | 33.4   | 37.8   | 26     | 2.6    | 0      | 0.2    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 8.9    | 10.2   | 8      | 1.8    | 0.1    | -      |
| U2 - cutoff = 0.85         | 9.3    | 10.6   | 8.1    | 1.7    | 0.1    | -      |
| U2 - cutoff = 0.95         | 9.2    | 10.6   | 8.1    | 1.9    | 0.1    | -      |
| U3 - cutoff = 0.75         | 9.4    | 10.4   | 7.6    | 1.5    | 0.1    | -      |
| U3 - cutoff = 0.85         | 9.8    | 10.8   | 7.6    | 1.6    | 0.1    | -      |
| U3 - cutoff = 0.95         | 9.6    | 10.8   | 7.8    | 1.8    | 0.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 2.2    | 4.4    | 2.3    | 0.7    | 0      | -      |
| U2 - cutoff = 0.85         | 2.4    | 4.5    | 2.4    | 0.7    | 0      | -      |
| U2 - cutoff = 0.95         | 0.8    | 4      | 2      | 0.8    | 0.1    | -      |
| U3 - cutoff = 0.75         | 2.4    | 4.5    | 2.2    | 0.6    | 0      | -      |
| U3 - cutoff = 0.85         | 2.6    | 4.6    | 2.2    | 0.6    | 0      | -      |
| U3 - cutoff = 0.95         | 0.8    | 4      | 2      | 0.8    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1071.5 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1100   |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1228.8 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1066.3 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1094.2 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1234.4 |
| Scenario 26                |        |        |        |        |        |        |
| DLT rate                   | 0.15   | 0.22   | 0.30   | 0.62   | 0.80   |        |
| Efficacy rate              | 0.10   | 0.20   | 0.35   | 0.45   | 0.55   |        |
| Utility 2                  | -0.1   | 0      | 0.1    | -0.2   | -0.2   |        |
| Utility 3                  | 40     | 43.2   | 49     | 42.2   | 41     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 17.1   | 28.2   | 23.8   | 2.8    | 0      | 28.1   |
| U2 - cutoff = 0.85         | 22.9   | 35.9   | 25.7   | 2.5    | 0      | 13     |
| U2 - cutoff = 0.95         | 24.8   | 42.3   | 26     | 2.5    | 0      | 4.3    |
| U3 - cutoff = 0.75         | 19.4   | 26.7   | 23.3   | 2.5    | 0      | 28.1   |
| U3 - cutoff = 0.85         | 23.9   | 37     | 23.4   | 2.2    | 0      | 13.5   |
| U3 - cutoff = 0.95         | 30.1   | 40.8   | 22.3   | 2.3    | 0      | 4.5    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 7.9    | 8.7    | 6.8    | 2      | 0.2    | -      |
| U2 - cutoff = 0.85         | 9.5    | 9.6    | 7      | 2.1    | 0.2    | -      |
| U2 - cutoff = 0.95         | 10.3   | 10.1   | 6.9    | 2.1    | 0.2    | -      |
| U3 - cutoff = 0.75         | 8.2    | 8.8    | 6.6    | 1.9    | 0.1    | -      |
| U3 - cutoff = 0.85         | 9.8    | 9.7    | 6.8    | 1.9    | 0.1    | -      |
| U3 - cutoff = 0.95         | 11     | 10.1   | 6.5    | 2      | 0.1    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.7    | 4.2    | 2      | 0.8    | 0      | -      |
| U2 - cutoff = 0.85         | 2.4    | 4.5    | 2.1    | 0.8    | 0      | -      |
| U2 - cutoff = 0.95         | 0.8    | 4.3    | 1.6    | 0.9    | 0.1    | -      |
| U3 - cutoff = 0.75         | 1.9    | 4.2    | 1.9    | 0.8    | 0      | -      |
| U3 - cutoff = 0.85         | 2.6    | 4.6    | 2      | 0.8    | 0      | -      |
| U3 - cutoff = 0.95         | 0.9    | 4.3    | 1.5    | 0.9    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 922.8  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1011.4 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1202.5 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 919.7  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1008.1 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1215.5 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.



Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 27                |        |        |        |        |        |        |
| DLT rate                   | 0.11   | 0.17   | 0.30   | 0.50   | 0.62   |        |
| Efficacy rate              | 0.10   | 0.20   | 0.40   | 0.42   | 0.45   |        |
| Utility 2                  | 0      | 0      | 0.1    | -0.1   | -0.2   |        |
| Utility 3                  | 41.6   | 45.2   | 52     | 45.2   | 42.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 12.9   | 24.1   | 40     | 6.6    | 0.1    | 16.3   |
| U2 - cutoff = 0.85         | 14     | 30.7   | 43.1   | 6.4    | 0.1    | 5.7    |
| U2 - cutoff = 0.95         | 17.1   | 32.7   | 41.2   | 7.4    | 0      | 1.7    |
| U3 - cutoff = 0.75         | 15.3   | 24.5   | 38.6   | 5.6    | 0      | 16     |
| U3 - cutoff = 0.85         | 17.2   | 30.9   | 41.5   | 4.9    | 0      | 5.5    |
| U3 - cutoff = 0.95         | 20.6   | 34.5   | 38.1   | 5.3    | 0.1    | 1.4    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 6.2    | 8.5    | 9.5    | 2.8    | 0.4    | -      |
| U2 - cutoff = 0.85         | 7.1    | 9.2    | 9.8    | 2.9    | 0.4    | -      |
| U2 - cutoff = 0.95         | 7.5    | 9.4    | 9.5    | 3.1    | 0.4    | -      |
| U3 - cutoff = 0.75         | 6.5    | 8.6    | 9.4    | 2.7    | 0.3    | -      |
| U3 - cutoff = 0.85         | 7.4    | 9.3    | 9.7    | 2.6    | 0.3    | -      |
| U3 - cutoff = 0.95         | 7.9    | 9.4    | 9.2    | 3      | 0.3    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.2    | 4.1    | 2.8    | 1.1    | 0.1    | -      |
| U2 - cutoff = 0.85         | 1.6    | 4.3    | 2.9    | 1.1    | 0.1    | -      |
| U2 - cutoff = 0.95         | 0.6    | 4      | 2.1    | 1.2    | 0.1    | -      |
| U3 - cutoff = 0.75         | 1.3    | 4.1    | 2.7    | 1      | 0.1    | -      |
| U3 - cutoff = 0.85         | 1.7    | 4.4    | 2.8    | 1      | 0.1    | -      |
| U3 - cutoff = 0.95         | 0.7    | 3.9    | 2.1    | 1.2    | 0.1    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1001.7 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1063.8 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1196.7 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1000.7 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1060.2 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1208.5 |
| Scenario 28                |        |        |        |        |        |        |
| DLT rate                   | 0.20   | 0.30   | 0.50   | 0.60   | 0.77   |        |
| Efficacy rate              | 0.05   | 0.30   | 0.32   | 0.35   | 0.40   |        |
| Utility 2                  | -0.1   | 0      | -0.2   | -0.2   | -0.4   |        |
| Utility 3                  | 35     | 46     | 39.2   | 37     | 33.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 9      | 24.5   | 2.7    | 0.2    | 0      | 63.6   |
| U2 - cutoff = 0.85         | 15.6   | 32.2   | 2.5    | 0.1    | 0      | 49.6   |
| U2 - cutoff = 0.95         | 28.2   | 40.6   | 4.3    | 0.1    | 0      | 26.7   |
| U3 - cutoff = 0.75         | 8.6    | 24.3   | 2.4    | 0.1    | 0      | 64.6   |
| U3 - cutoff = 0.85         | 15.9   | 31.8   | 2.6    | 0      | 0      | 49.7   |
| U3 - cutoff = 0.95         | 29.7   | 40.2   | 3.3    | 0      | 0      | 26.8   |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 7.9    | 8.8    | 2.5    | 0.6    | 0      | -      |
| U2 - cutoff = 0.85         | 10.6   | 10     | 2.6    | 0.6    | 0      | -      |
| U2 - cutoff = 0.95         | 13.3   | 10.8   | 2.7    | 0.6    | 0      | -      |
| U3 - cutoff = 0.75         | 8      | 8.8    | 2.4    | 0.6    | 0      | -      |
| U3 - cutoff = 0.85         | 10.7   | 9.9    | 2.6    | 0.5    | 0      | -      |
| U3 - cutoff = 0.95         | 13.6   | 10.7   | 2.6    | 0.6    | 0      | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.5    | 4      | 0.8    | 0.2    | 0      | -      |
| U2 - cutoff = 0.85         | 2.7    | 4.5    | 0.8    | 0.3    | 0      | -      |
| U2 - cutoff = 0.95         | 0.9    | 4.6    | 0.9    | 0.3    | 0      | -      |
| U3 - cutoff = 0.75         | 1.6    | 4      | 0.7    | 0.2    | 0      | -      |
| U3 - cutoff = 0.85         | 2.7    | 4.4    | 0.8    | 0.2    | 0      | -      |
| U3 - cutoff = 0.95         | 0.9    | 4.5    | 0.9    | 0.3    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 730.2  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 854    |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1147.8 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 726.2  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 848.9  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1155.5 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 29                |        |        |        |        |        |        |
| DLT rate                   | 0.19   | 0.30   | 0.56   | 0.74   | 0.86   |        |
| Efficacy rate              | 0.40   | 0.40   | 0.40   | 0.40   | 0.40   |        |
| Utility 2                  | 0.2    | 0.1    | -0.2   | -0.3   | -0.5   |        |
| Utility 3                  | 56.4   | 52     | 41.6   | 34.4   | 29.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 67.5   | 26.1   | 1.8    | 0      | 0      | 4.6    |
| U2 - cutoff = 0.85         | 69.2   | 26.2   | 1.8    | 0      | 0      | 2.8    |
| U2 - cutoff = 0.95         | 61     | 33.8   | 2.4    | 0      | 0      | 2.8    |
| U3 - cutoff = 0.75         | 69.9   | 24.5   | 0.9    | 0      | 0      | 4.7    |
| U3 - cutoff = 0.85         | 72.3   | 24     | 0.9    | 0      | 0      | 2.8    |
| U3 - cutoff = 0.95         | 64     | 31.9   | 1.5    | 0      | 0      | 2.6    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 18.9   | 8.5    | 1.5    | 0.2    | 0      | -      |
| U2 - cutoff = 0.85         | 19.1   | 8.6    | 1.5    | 0.2    | 0      | -      |
| U2 - cutoff = 0.95         | 18     | 9.5    | 1.8    | 0.2    | 0      | -      |
| U3 - cutoff = 0.75         | 19.8   | 8.1    | 1.1    | 0.1    | 0      | -      |
| U3 - cutoff = 0.85         | 20     | 8.2    | 1.2    | 0.1    | 0      | -      |
| U3 - cutoff = 0.95         | 18.7   | 9.3    | 1.3    | 0.2    | 0      | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.7    | 3.1    | 0.5    | 0.1    | 0      | -      |
| U2 - cutoff = 0.85         | 4.7    | 3.2    | 0.5    | 0.1    | 0      | -      |
| U2 - cutoff = 0.95         | 0.8    | 3.3    | 0.8    | 0.1    | 0      | -      |
| U3 - cutoff = 0.75         | 5      | 3      | 0.3    | 0.1    | 0      | -      |
| U3 - cutoff = 0.85         | 5.1    | 3.1    | 0.4    | 0      | 0      | -      |
| U3 - cutoff = 0.95         | 0.9    | 3.2    | 0.6    | 0.1    | 0      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1150.2 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1160.6 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1352.1 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1146.2 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1157.8 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1371.4 |
|                            |        |        |        |        |        |        |
| Scenario 30                |        |        |        |        |        |        |
| DLT rate                   | 0.05   | 0.08   | 0.17   | 0.27   | 0.54   |        |
| Efficacy rate              | 0.12   | 0.15   | 0.20   | 0.40   | 0.50   |        |
| Utility 2                  | 0.1    | 0.1    | 0      | 0.1    | 0      |        |
| Utility 3                  | 45.2   | 45.8   | 45.2   | 53.2   | 48.4   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 15.6   | 12.3   | 15.4   | 45.4   | 8.7    | 2.6    |
| U2 - cutoff = 0.85         | 15.9   | 13.6   | 16.1   | 45.3   | 8.6    | 0.5    |
| U2 - cutoff = 0.95         | 17.1   | 15.3   | 16.1   | 41.2   | 10.4   | 0      |
| U3 - cutoff = 0.75         | 21.2   | 14.5   | 14.5   | 39.8   | 7.2    | 2.8    |
| U3 - cutoff = 0.85         | 22.2   | 15.3   | 15.2   | 39.9   | 6.9    | 0.5    |
| U3 - cutoff = 0.95         | 21.9   | 17.8   | 16.8   | 35.2   | 8.4    | 0      |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.4    | 5.5    | 6.4    | 10.1   | 3.1    | -      |
| U2 - cutoff = 0.85         | 4.5    | 5.6    | 6.6    | 10.1   | 3.2    | -      |
| U2 - cutoff = 0.95         | 4.6    | 5.7    | 6.6    | 9.8    | 3.3    | -      |
| U3 - cutoff = 0.75         | 4.6    | 5.6    | 6.4    | 10     | 2.8    | -      |
| U3 - cutoff = 0.85         | 4.9    | 5.7    | 6.5    | 10.1   | 2.8    | -      |
| U3 - cutoff = 0.95         | 5      | 5.8    | 6.6    | 9.6    | 2.9    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.6    | 3.2    | 2.1    | 3.4    | 1      | -      |
| U2 - cutoff = 0.85         | 0.6    | 3.2    | 2.1    | 3.3    | 1      | -      |
| U2 - cutoff = 0.95         | 0.3    | 3      | 2      | 2.3    | 1.1    | -      |
| U3 - cutoff = 0.75         | 0.7    | 3.3    | 2.1    | 3.4    | 0.9    | -      |
| U3 - cutoff = 0.85         | 0.8    | 3.3    | 2.1    | 3.4    | 0.9    | -      |
| U3 - cutoff = 0.95         | 0.3    | 3      | 2      | 2.3    | 0.9    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1062.7 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1075.7 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1177.3 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1057.1 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1072.3 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1183.8 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 31                |        |        |        |        |        |        |
| DLT rate                   | 0.04   | 0.06   | 0.10   | 0.20   | 0.60   |        |
| Efficacy rate              | 0.20   | 0.50   | 0.50   | 0.50   | 0.50   |        |
| Utility 2                  | 0.2    | 0.4    | 0.4    | 0.3    | -0.1   |        |
| Utility 3                  | 50.4   | 67.6   | 66     | 62     | 46     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 2      | 44.9   | 31     | 20     | 1.9    | 0.2    |
| U2 - cutoff = 0.85         | 2.3    | 45.1   | 31.3   | 19.3   | 2      | 0      |
| U2 - cutoff = 0.95         | 2.2    | 39.9   | 31.7   | 23.4   | 2.9    | 0      |
| U3 - cutoff = 0.75         | 3.9    | 49.7   | 28.6   | 16.7   | 0.9    | 0.2    |
| U3 - cutoff = 0.85         | 3.8    | 50.8   | 27.9   | 16.6   | 0.9    | 0      |
| U3 - cutoff = 0.95         | 4.1    | 45.8   | 30.4   | 18.2   | 1.6    | 0      |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 3.9    | 14.3   | 7.5    | 3.7    | 0.6    | -      |
| U2 - cutoff = 0.85         | 3.9    | 14.3   | 7.6    | 3.7    | 0.5    | -      |
| U2 - cutoff = 0.95         | 3.9    | 13.6   | 7.8    | 4.1    | 0.6    | -      |
| U3 - cutoff = 0.75         | 4.1    | 15.1   | 7.1    | 3.3    | 0.4    | -      |
| U3 - cutoff = 0.85         | 4.1    | 15.3   | 7      | 3.2    | 0.4    | -      |
| U3 - cutoff = 0.95         | 4      | 14.5   | 7.5    | 3.6    | 0.5    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.3    | 5.4    | 2.3    | 1.1    | 0.2    | -      |
| U2 - cutoff = 0.85         | 0.3    | 5.4    | 2.3    | 1.1    | 0.2    | -      |
| U2 - cutoff = 0.95         | 0.1    | 2.7    | 1.7    | 1      | 0.3    | -      |
| U3 - cutoff = 0.75         | 0.4    | 5.7    | 2.2    | 1      | 0.1    | -      |
| U3 - cutoff = 0.85         | 0.4    | 5.8    | 2.2    | 1      | 0.1    | -      |
| U3 - cutoff = 0.95         | 0.1    | 2.8    | 1.6    | 0.9    | 0.2    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1137   |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1139   |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1340   |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1134.3 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1135.4 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1356.6 |
| Scenario 32                |        |        |        |        |        |        |
| DLT rate                   | 0.02   | 0.05   | 0.11   | 0.30   | 0.36   |        |
| Efficacy rate              | 0.08   | 0.15   | 0.30   | 0.30   | 0.30   |        |
| Utility 2                  | 0.1    | 0.1    | 0.2    | 0      | -0.1   |        |
| Utility 3                  | 44     | 47     | 53.6   | 46     | 43.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 10.7   | 14     | 34.2   | 31.2   | 9.3    | 0.6    |
| U2 - cutoff = 0.85         | 11.6   | 13.7   | 33.9   | 31.5   | 9.1    | 0.2    |
| U2 - cutoff = 0.95         | 11.4   | 15.2   | 31.3   | 31.2   | 10.9   | 0      |
| U3 - cutoff = 0.75         | 14.9   | 17.6   | 34.2   | 26.4   | 6.3    | 0.6    |
| U3 - cutoff = 0.85         | 15.5   | 17.9   | 34     | 27.1   | 5.4    | 0.1    |
| U3 - cutoff = 0.95         | 15.6   | 19.6   | 31.7   | 26.6   | 6.5    | 0      |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 3.5    | 4.6    | 8.8    | 8.8    | 4.1    | -      |
| U2 - cutoff = 0.85         | 3.5    | 4.5    | 8.9    | 8.9    | 4.2    | -      |
| U2 - cutoff = 0.95         | 3.5    | 4.6    | 8.7    | 9      | 4.1    | -      |
| U3 - cutoff = 0.75         | 3.6    | 4.7    | 9.3    | 8.7    | 3.5    | -      |
| U3 - cutoff = 0.85         | 3.6    | 4.7    | 9.4    | 8.7    | 3.6    | -      |
| U3 - cutoff = 0.95         | 3.6    | 4.9    | 9.1    | 8.8    | 3.6    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.2    | 3      | 2.9    | 3.3    | 1.4    | -      |
| U2 - cutoff = 0.85         | 0.2    | 2.9    | 2.9    | 3.4    | 1.4    | -      |
| U2 - cutoff = 0.95         | 0.1    | 2.7    | 2      | 2.9    | 1.2    | -      |
| U3 - cutoff = 0.75         | 0.2    | 3      | 3.1    | 3.3    | 1.2    | -      |
| U3 - cutoff = 0.85         | 0.3    | 3      | 3.1    | 3.3    | 1.2    | -      |
| U3 - cutoff = 0.95         | 0.1    | 2.8    | 2.1    | 2.8    | 1      | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1048.3 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1054.2 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1153.7 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1045.1 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1050.1 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1161.5 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 33                |        |        |        |        |        |        |
| DLT rate                   | 0.01   | 0.02   | 0.09   | 0.32   | 0.38   |        |
| Efficacy rate              | 0.05   | 0.08   | 0.20   | 0.40   | 0.50   |        |
| Utility 2                  | 0      | 0.1    | 0.1    | 0.1    | 0.1    |        |
| Utility 3                  | 42.6   | 44     | 48.4   | 51.2   | 54.8   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 8.6    | 6.3    | 14.2   | 50.7   | 19.8   | 0.4    |
| U2 - cutoff = 0.85         | 7.8    | 6.1    | 13.6   | 51.8   | 20.6   | 0.1    |
| U2 - cutoff = 0.95         | 8.5    | 6.2    | 11.6   | 49.2   | 24.5   | 0      |
| U3 - cutoff = 0.75         | 12.1   | 7      | 13.4   | 49.2   | 17.9   | 0.4    |
| U3 - cutoff = 0.85         | 12     | 6.6    | 13.3   | 50.3   | 17.7   | 0.1    |
| U3 - cutoff = 0.95         | 11.9   | 8.9    | 12.3   | 46.3   | 20.5   | 0      |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 3.3    | 3.8    | 6.2    | 11.3   | 5.2    | -      |
| U2 - cutoff = 0.85         | 3.3    | 3.8    | 6.3    | 11.2   | 5.4    | -      |
| U2 - cutoff = 0.95         | 3.3    | 3.9    | 6.2    | 10.9   | 5.6    | -      |
| U3 - cutoff = 0.75         | 3.4    | 3.8    | 6.3    | 11.4   | 4.8    | -      |
| U3 - cutoff = 0.85         | 3.5    | 3.9    | 6.4    | 11.5   | 4.8    | -      |
| U3 - cutoff = 0.95         | 3.4    | 4      | 6.4    | 11     | 5.2    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.1    | 2.7    | 1.8    | 4      | 1.4    | -      |
| U2 - cutoff = 0.85         | 0.1    | 2.7    | 1.9    | 4      | 1.5    | -      |
| U2 - cutoff = 0.95         | 0.1    | 2.7    | 1.5    | 2.8    | 1.2    | -      |
| U3 - cutoff = 0.75         | 0.2    | 2.8    | 1.9    | 4      | 1.3    | -      |
| U3 - cutoff = 0.85         | 0.2    | 2.8    | 1.9    | 4      | 1.3    | -      |
| U3 - cutoff = 0.95         | 0.1    | 2.7    | 1.6    | 2.8    | 1.1    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1079.4 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1087.4 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1190.2 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1077.4 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1084.1 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1195.5 |
|                            |        |        |        |        |        |        |
| Scenario 34                |        |        |        |        |        |        |
| DLT rate                   | 0.05   | 0.13   | 0.30   | 0.36   | 0.42   |        |
| Efficacy rate              | 0.05   | 0.10   | 0.30   | 0.32   | 0.35   |        |
| Utility 2                  | 0      | 0      | 0      | 0      | -0.1   |        |
| Utility 3                  | 41     | 40.8   | 46     | 44.8   | 44.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 11.3   | 13.8   | 46     | 14.2   | 2.4    | 12.3   |
| U2 - cutoff = 0.85         | 12.6   | 15.9   | 49.3   | 14.8   | 2.4    | 5      |
| U2 - cutoff = 0.95         | 15.1   | 19.5   | 47.2   | 16.1   | 2      | 0.2    |
| U3 - cutoff = 0.75         | 14     | 15     | 43.2   | 13.4   | 1.8    | 12.6   |
| U3 - cutoff = 0.85         | 16.3   | 18.9   | 44.3   | 13.5   | 2      | 5      |
| U3 - cutoff = 0.95         | 17.7   | 22.2   | 43.6   | 14.7   | 1.5    | 0.3    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.7    | 6.5    | 9.8    | 5.4    | 1.5    | -      |
| U2 - cutoff = 0.85         | 5.4    | 7      | 10.3   | 5.4    | 1.5    | -      |
| U2 - cutoff = 0.95         | 5.7    | 7.3    | 9.9    | 5.5    | 1.5    | -      |
| U3 - cutoff = 0.75         | 5      | 6.6    | 9.9    | 5.2    | 1.3    | -      |
| U3 - cutoff = 0.85         | 5.7    | 7.2    | 10.2   | 5.3    | 1.3    | -      |
| U3 - cutoff = 0.95         | 6.2    | 7.2    | 9.8    | 5.4    | 1.3    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.7    | 3.6    | 3.1    | 1.9    | 0.5    | -      |
| U2 - cutoff = 0.85         | 1      | 3.8    | 3.2    | 2      | 0.5    | -      |
| U2 - cutoff = 0.95         | 0.6    | 3.7    | 2.3    | 1.9    | 0.5    | -      |
| U3 - cutoff = 0.75         | 0.9    | 3.7    | 3.1    | 1.9    | 0.5    | -      |
| U3 - cutoff = 0.85         | 1.2    | 3.9    | 3.2    | 1.9    | 0.5    | -      |
| U3 - cutoff = 0.95         | 0.6    | 3.7    | 2.2    | 1.8    | 0.4    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 988.3  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1044.9 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1163.8 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 986.1  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1041.7 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1174.2 |

Priors: toxicity alpha Beta(0.15, 0.85); efficacy beta-binomial Beta(0.5, 0.5); efficacy alpha Beta(0.15, 0.85).  
Cutoffs 0.75 and 0.85 use August 10 data with efficacy threshold = 0.20; cutoff 0.95 uses the specified August 3 source.



Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 35                |        |        |        |        |        |        |
| DLT rate                   | 0.03   | 0.09   | 0.30   | 0.40   | 0.45   |        |
| Efficacy rate              | 0.20   | 0.45   | 0.45   | 0.45   | 0.45   |        |
| Utility 2                  | 0.2    | 0.4    | 0.1    | 0.1    | 0      |        |
| Utility 3                  | 50.8   | 63.4   | 55     | 51     | 49     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 6      | 48.2   | 36.7   | 7.8    | 1      | 0.3    |
| U2 - cutoff = 0.85         | 6.4    | 47.8   | 36.5   | 8.2    | 1      | 0.1    |
| U2 - cutoff = 0.95         | 7.4    | 41.7   | 38.4   | 10.9   | 1.6    | 0      |
| U3 - cutoff = 0.75         | 9.3    | 53.2   | 30.2   | 6.2    | 0.8    | 0.3    |
| U3 - cutoff = 0.85         | 9.1    | 53.1   | 30.2   | 6.8    | 0.7    | 0.1    |
| U3 - cutoff = 0.95         | 9.7    | 48.8   | 32.7   | 8.2    | 0.6    | 0      |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 4.4    | 13.8   | 8.6    | 2.7    | 0.5    | -      |
| U2 - cutoff = 0.85         | 4.3    | 13.7   | 8.7    | 2.8    | 0.5    | -      |
| U2 - cutoff = 0.95         | 4.4    | 13.1   | 8.9    | 3      | 0.6    | -      |
| U3 - cutoff = 0.75         | 4.7    | 14.6   | 7.9    | 2.3    | 0.4    | -      |
| U3 - cutoff = 0.85         | 4.7    | 14.7   | 7.9    | 2.3    | 0.4    | -      |
| U3 - cutoff = 0.95         | 4.5    | 14.1   | 8.2    | 2.7    | 0.4    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 0.5    | 5.3    | 2.4    | 0.9    | 0.2    | -      |
| U2 - cutoff = 0.85         | 0.5    | 5.3    | 2.5    | 0.9    | 0.1    | -      |
| U2 - cutoff = 0.95         | 0.2    | 3.1    | 2.2    | 1      | 0.2    | -      |
| U3 - cutoff = 0.75         | 0.6    | 5.7    | 2.2    | 0.8    | 0.1    | -      |
| U3 - cutoff = 0.85         | 0.6    | 5.7    | 2.2    | 0.8    | 0.1    | -      |
| U3 - cutoff = 0.95         | 0.2    | 3.1    | 2      | 0.9    | 0.1    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 1141.1 |
| U2 - cutoff = 0.85         |        |        |        |        |        | 1143.8 |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1290.9 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 1136.2 |
| U3 - cutoff = 0.85         |        |        |        |        |        | 1136.5 |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1312   |
| Scenario 36                |        |        |        |        |        |        |
| DLT rate                   | 0.13   | 0.29   | 0.35   | 0.40   | 0.43   |        |
| Efficacy rate              | 0.05   | 0.20   | 0.22   | 0.30   | 0.35   |        |
| Utility 2                  | -0.1   | -0.1   | -0.1   | -0.1   | -0.1   |        |
| Utility 3                  | 37.8   | 40.4   | 39.2   | 42     | 43.8   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 10.7   | 28.5   | 10.4   | 3.1    | 0.7    | 46.6   |
| U2 - cutoff = 0.85         | 15.1   | 39.7   | 12.8   | 3.5    | 0.4    | 28.5   |
| U2 - cutoff = 0.95         | 24.6   | 48.6   | 11.7   | 3.2    | 0.9    | 11     |
| U3 - cutoff = 0.75         | 13.2   | 28.3   | 8.6    | 2.5    | 0.5    | 46.9   |
| U3 - cutoff = 0.85         | 17.6   | 40.2   | 10.3   | 2.7    | 0.3    | 28.9   |
| U3 - cutoff = 0.95         | 28.3   | 46     | 10.2   | 3.1    | 0.5    | 12     |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 6.8    | 8.7    | 4.5    | 2.4    | 0.5    | -      |
| U2 - cutoff = 0.85         | 8.8    | 10.2   | 4.8    | 2.4    | 0.5    | -      |
| U2 - cutoff = 0.95         | 10.4   | 11.1   | 4.8    | 2.4    | 0.6    | -      |
| U3 - cutoff = 0.75         | 7      | 8.8    | 4.3    | 2.3    | 0.5    | -      |
| U3 - cutoff = 0.85         | 9.1    | 10.3   | 4.6    | 2.3    | 0.4    | -      |
| U3 - cutoff = 0.95         | 10.9   | 10.9   | 4.5    | 2.3    | 0.5    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.2    | 4.3    | 1.4    | 0.9    | 0.2    | -      |
| U2 - cutoff = 0.85         | 2.1    | 5      | 1.5    | 1      | 0.2    | -      |
| U2 - cutoff = 0.95         | 0.8    | 5.1    | 1.4    | 0.9    | 0.2    | -      |
| U3 - cutoff = 0.75         | 1.3    | 4.4    | 1.4    | 0.9    | 0.2    | -      |
| U3 - cutoff = 0.85         | 2.2    | 5      | 1.5    | 0.9    | 0.2    | -      |
| U3 - cutoff = 0.95         | 0.9    | 5      | 1.3    | 0.8    | 0.2    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 810.1  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 935.3  |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1145.5 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 804.9  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 930.9  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1152.3 |

Phase I/II Operating Characteristics

Non-TITE efficacy cutoff comparison - IPDE alpha = 0.6 - allocation = one-stage - N = 30

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 37                |        |        |        |        |        |        |
| DLT rate                   | 0.08   | 0.30   | 0.40   | 0.45   | 0.50   |        |
| Efficacy rate              | 0.05   | 0.20   | 0.20   | 0.20   | 0.20   |        |
| Utility 2                  | 0      | -0.1   | -0.2   | -0.2   | -0.3   |        |
| Utility 3                  | 39.8   | 40     | 36     | 34     | 32     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 10.7   | 35.2   | 10.1   | 1      | 0      | 43     |
| U2 - cutoff = 0.85         | 15.1   | 48.9   | 11.2   | 1      | 0      | 23.8   |
| U2 - cutoff = 0.95         | 24.3   | 57.7   | 12.3   | 1.4    | 0      | 4.3    |
| U3 - cutoff = 0.75         | 12     | 34.5   | 9.3    | 1      | 0.1    | 43.1   |
| U3 - cutoff = 0.85         | 17.5   | 47.9   | 10.1   | 0.9    | 0      | 23.6   |
| U3 - cutoff = 0.95         | 28.4   | 54.8   | 10.4   | 1.2    | 0      | 5.1    |
| Mean administrations       |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 6.7    | 10.1   | 4.6    | 2      | 0.4    | -      |
| U2 - cutoff = 0.85         | 8.6    | 11.7   | 4.9    | 2.1    | 0.4    | -      |
| U2 - cutoff = 0.95         | 9.8    | 12.3   | 5      | 2.1    | 0.4    | -      |
| U3 - cutoff = 0.75         | 6.9    | 10.1   | 4.5    | 1.9    | 0.4    | -      |
| U3 - cutoff = 0.85         | 8.9    | 11.8   | 4.7    | 2      | 0.4    | -      |
| U3 - cutoff = 0.95         | 10.4   | 12.1   | 4.8    | 2      | 0.4    | -      |
| Mean IPDE administrations  |        |        |        |        |        |        |
| U2 - cutoff = 0.75         | 1.2    | 5      | 1.4    | 0.8    | 0.1    | -      |
| U2 - cutoff = 0.85         | 2      | 5.6    | 1.5    | 0.9    | 0.1    | -      |
| U2 - cutoff = 0.95         | 1      | 5.4    | 1.5    | 0.8    | 0.2    | -      |
| U3 - cutoff = 0.75         | 1.3    | 5      | 1.4    | 0.8    | 0.1    | -      |
| U3 - cutoff = 0.85         | 2.2    | 5.7    | 1.5    | 0.8    | 0.1    | -      |
| U3 - cutoff = 0.95         | 1.1    | 5.3    | 1.5    | 0.8    | 0.1    | -      |
| Mean trial duration (days) |        |        |        |        |        | Days   |
| U2 - cutoff = 0.75         |        |        |        |        |        | 838.8  |
| U2 - cutoff = 0.85         |        |        |        |        |        | 960    |
| U2 - cutoff = 0.95         |        |        |        |        |        | 1145.7 |
| U3 - cutoff = 0.75         |        |        |        |        |        | 837.3  |
| U3 - cutoff = 0.85         |        |        |        |        |        | 959.7  |
| U3 - cutoff = 0.95         |        |        |        |        |        | 1152.3 |